# Timing Search†

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**Abstract**

Why do industries fluctuate at different frequencies, though they share the same aggregate shocks? I develop a timing-search model in which agents choose when to enter, but matching frictions make rapid entry costly: firms cluster their entry in time, but they also congest a shared entry process. In equilibrium, the industry never settles at the efficient scale but constantly overshoots around it, leading to an endogenous cycle. The cycle frequency is pinned by industry-level primitives, and the closed form $\omega = \sqrt{\varphi/\mu}$ is set by profit curvature $\varphi$ and matching friction $\mu$. Consistent with the model, U.S. manufacturing industries cycle at different frequencies, ordered by market structure. Permitting reform and cluster subsidies change only the cycle's speed, not its welfare cost from the congestion externality; the efficient remedy taxes entry, while capacity-expansion subsidies do the opposite, enlarging the cycle and raising it.

**Keywords**: Timing search, cycle frequency, market structure, temporal indifference, scale externality.

**JEL codes**: D83, E32, L16, E22.

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# 1 Introduction

Why do different industries fluctuate at different frequencies? In U.S. data, industries cycle at strikingly different lengths: their production swings recur roughly every 3.0 to 4.7 years, and waves of new-establishment entry every 2.4 to nearly 10 years. Standard business-cycle frameworks derive cycle properties from a common shock process (Kydland and Prescott, 1982) or policy rule, objects common across industries, so this cross-industry variation points to an industry-specific source. A natural question remains open: how an industry's market structure shapes its cycle's frequency and amplitude.

This paper addresses the gap by introducing *timing search* equilibrium, which brings the directed-search framework (Moen, 1997; Wright et al., 2021) to the choice of *when* to enter: agents direct their search across entry dates rather than across contemporaneous submarkets, and equilibrium equalizes the value of entry across dates, the temporal counterpart of the equal-value condition in the wage-posting model of Burdett and Mortensen (1998). The novel ingredient is *time* as the submarket index. Adjacency between submarket indices such as wage is a modeling convention, while adjacency between dates has a physical implication, because time flows in one direction. Each date inherits the industry that earlier entry built—an agent can postpone entry but never move it earlier—and neighboring dates compete for the same permits, specialist labor, and equipment, so rapid entry congests these shared inputs. The resulting friction falls on adjustment across neighboring submarkets, and it is the key factor generating the endogenous cycle. Semiconductors are the textbook case: firms rush to build fabrication capacity when chips are profitable, but they share a slow pipeline of permits, specialist crews, and equipment, so capacity overshoots into a glut, the correction overshoots into a shortage, and the industry cycles. How fast it cycles depends on the industry's own economics, not on common shocks, which is why different industries cycle at different lengths.

The contributions of this paper are threefold. First, I show that this friction makes industries cycle, at a frequency available in closed form (Theorem 1, Section 3). The equilibrium is governed by an indifference condition across entry dates, the *ironing condition*: no agent can profit by retiming entry, so the value of entering is equalized across dates. As an industry approaches optimal scale, this condition keeps the inflow at a rate that cannot drop to zero exactly on arrival without leaving a profitable entry date unfilled, so the industry overshoots optimal scale and circles it rather than settling there. The resulting cycle frequency is

$$\omega = \sqrt{\varphi/\mu}, \tag{1}$$

where $\varphi$ is the curvature of profits around optimal scale and $\mu$ is the entry-matching friction, both market-structure primitives rather than properties of the shock process. The underlying equilibrium concept, *time-indifference equilibrium*, is the temporal analog of the equal-value conditions in Hotelling (1931) and Burdett and Mortensen (1998). In the timing-search model, the cycle period depends on market structure alone, not on cycle amplitude; deeper recessions therefore recover at proportionally faster speeds in the same time as shallow ones, matching the plucking pattern of Friedman (1993) in U.S. industrial production over 1972–2024.

Second, I take the model's prediction that cycle frequency is pinned by market structure, rather than by the shock process, to U.S. industry data (Section 4). Across 15 manufacturing 3-digit industries, the spectral peak frequency varies systematically with the Ellison and Glaeser (1997) geographic concentration index, an empirical proxy for profit curvature. Industries with stronger agglomeration have higher spectral peak frequencies. A historical-geography IV built from 1947 state-level specialization, net of one-county dominance, has a robust first-stage statistic above 10 and yields a cross-industry slope consistent with the structural prediction. Primitive Kim and County Business Patterns instruments give the same sign, as does an independent BDS subsector cross-section.

This evidence is consistent with a profit-curvature channel. Across manufacturing, entry inputs such as permits, construction crews, and equipment supply chains flow relatively freely. The matching friction therefore varies little across 3-digit industries, and the cross-industry signal in cycle frequency comes from differences in the profit landscape. Three additional diagnostics are consistent with the mechanism but rest on weaker identification. BDS net-entry rates show oscillatory features outside the AR(1) class in 10 of 19 sectors. Manufacturing and oil investment oscillate at industry-specific frequencies after large historical shocks. After COVID-19, 12 of 19 BDS sectors overshoot above their pre-pandemic trend by at least one pre-COVID standard deviation. Among potential alternatives (the models of AR(1) RBC, New Keynesian AR(2), Beaudry et al. (2020), and multi-sector RBC with input-output linkages), each captures part of the joint pattern but not the cross-industry slope at observed magnitudes. Reproducing the full pattern requires a mechanism that maps industry-level primitives into industry-level frequencies (Online Appendix F).

Third, I build an industrial-policy taxonomy on the model's central prediction (Theorem 3, Section 5.3): each industry cycles at a frequency fixed by its market structure $(\varphi, \mu)$, while the welfare cost of that cycle is governed by a disjoint primitive, the entry-cost level $\bar{\kappa}$. This frequency-welfare separation sorts entry-side policies into those that move the welfare cost and those that only change the cycle's speed. Two natural classes of policy therefore fall on the timing-only side. Streamlining entry matching, such as faster permitting, and strengthening agglomeration, such as cluster subsidies, change cycle speed without affecting welfare cost. To leading order, only policies that raise the entry cost (higher $\bar{\kappa}$) move the mean welfare cost; stabilizing the entry cost (lower $\sigma_\kappa$) reduces the cycle's variability but not its mean (the volatility enters only at second order). The taxonomy distinguishes welfare-improving policy from policy that visibly changes cycle dynamics but leaves welfare unchanged. This distinction is relevant to the recent industrial-policy turn: the U.S. CHIPS and Science Act, the Inflation Reduction Act, and the EU Net-Zero Industry Act (Juhász et al., 2024). The framework asks which entry-side instruments move welfare and which reshape cycle dynamics. Under the per-firm welfare metric the model adopts, easing the entry bottleneck does not smooth the cycle but enlarges it: the industry keeps cycling at its structural frequency regardless, and the separation holds that frequency fixed. CHIPS- and IRA-style "capacity expansion" subsidies that lower the entry cost (lower $\bar{\kappa}$) therefore *raise* the welfare cost of cycles by enlarging their amplitude (Section 5.3). Section 5.2 develops the efficiency reading behind this result: the cyclical cost reflects a congestion externality in the timing of entry, for which a rebated tax is a natural corrective instrument.

**Related literature** This paper builds on four literatures.

*Structural and endogenous cycles.* The structural-cycle tradition is anchored by Friedman (1993), who reads recessions as plucks below an unobserved natural ceiling rather than as symmetric stochastic deviations. The deeper-recession-faster-recovery pattern predicted by that view is revisited empirically by Bordo and Haubrich (2017) and Dupraz et al. (2019). It emerges here as a corollary of the depth-independent period (Section 3.4). The closest formal macroeconomic antecedent is Beaudry et al. (2020), who show that financial frictions can generate endogenous boom-bust dynamics through a stochastic limit cycle (extended in Beaudry et al. (2024)). Beaudry et al. (2020) obtain a true limit cycle from a Hopf bifurcation, so the existence of cycles is a structurally stable prediction of their framework. The timing-search cycle's amplitude is likewise pinned down: it is a structural quantity, set by the entry-cost level $\bar{\kappa}$ and recovered from the data (Propositions 3–4).

An interaction-based tradition generates aggregate fluctuations by coordinating discrete adjustments: threshold rules amplify idiosyncratic shocks into aggregate fluctuations (Nirei, 2006), strategic complementarity bunches lumpy investments into aggregate bursts (Nirei, 2015), and the same logic produces repricing avalanches (Nirei and Scheinkman, 2024). Timing search runs on the opposite force: congestion across neighboring entry dates pushes agents to adjust apart rather than together. The two forces leave different fingerprints. Complementarity clusters adjustment into spikes and governs the size of aggregate fluctuations while leaving their timing unrestricted; congestion spreads entry smoothly over time and, joined with the profit-curvature restoring force, pins each industry's cycle at the closed-form frequency $\omega_i = \sqrt{\varphi_i/\mu_i}$. These are the fingerprints the empirical sections measure: smooth, sign-reversing entry rates and industry-specific spectral peaks. The complementarity route can itself generate recurrence: Shleifer (1986) obtains implementation cycles from the timing of adjustment, but through demand complementarity that synchronizes firms into a single economy-wide cycle, whereas congestion spaces entry and ties the frequency to each industry's own $(\varphi_i, \mu_i)$.

Relatedly, the timing-search route yields a closed-form mapping from market-structure primitives to cycle frequency and a frequency-welfare separation theorem. The animal-spirits and self-fulfilling-prophecies tradition of Farmer and Guo (1994) and Farmer (1999) generates endogenous fluctuations from indeterminacy and belief shocks. Here, by contrast, cycle frequency is pinned by market-structure primitives. In the stochastic equilibrium, cycle amplitude is pinned by the entry-cost level (stationary amplitude $\sigma_x = \sqrt{\bar{b}/\varphi}$) rather than by self-fulfilling beliefs. The frequency-vs-welfare disjointness is reserved for Theorem 3: only there does $\varphi$ cancel from the welfare formula $\mathbb{E}[\mathcal{L}] = \bar{b}$, while it remains a determinant of amplitude. Beaudry and Portier (2006) document stock prices leading TFP; Grandmont (1985), Azariadis (1981), Benhabib and Farmer (1994), and Goodwin (1967) provide earlier endogenous-cycle precedents. The closest timing-based antecedent is Jovanovic (2009), who shows how investment timing can generate endogenous business cycles. The present paper shifts the object from the option value of investment timing to the equilibrium distribution of entry dates, which yields a closed-form mapping from market-structure primitives to cycle frequency.

*Search and matching frictions.* Timing search is in the spirit of Stigler (1961), McCall (1970), and especially Burdett and Mortensen (1998). My paper is a temporal counterpart in which agents choose among dates rather than among contemporaneous alternatives. The ironing condition is analogous to the equal-value condition in Burdett and Mortensen (1998) (Online Appendix A.1). The micro-foundation $\mu = (1-\alpha)/[\alpha(\delta n_p)^2]$ further connects to the Diamond-Mortensen-Pissarides literature (Diamond, 1982; Mortensen and Pissarides, 1994; Pissarides, 2000; Petrongolo and Pissarides, 2001; Shimer, 2005). Matching frictions in entry generate equilibrium business cycles in the same way DMP matching frictions generate equilibrium unemployment. The friction on the *rate* of entry has a direct antecedent in Caballero and Hammour (1994), whose convex creation costs smooth entry over the cycle; the same cost on $\dot{n}$ operates here, but combined with the profit-curvature restoring force and the equal-value condition it generates the cycle rather than damping the response to shocks.

*Industry dynamics.* The model isolates a timing-search channel within an endogenous entry environment. It is complementary to the richer heterogeneity and selection mechanisms of Hopenhayn (1992) and Hopenhayn and Rogerson (1993). The entry-wave dynamics generated by the ironing condition also complement Jovanovic (1982)'s learning-based exit explanation, while Davis and Haltiwanger (1992) document the entry/exit empirics this literature studies. A separate strand on aggregate fluctuations from disaggregated origins (Gabaix, 2011; Acemoglu et al., 2012) provides the natural alternative the paper tests against in Online Appendix F.2.

*Agglomeration.* The scale-congestion logic of Krugman (1991), Duranton and Puga (2004), Rosenthal and Strange (2004), Ellison and Glaeser (1997), and Ellison et al. (2010) is transposed from space to time in my paper. Combined with timing frictions, it generates temporal patterns (cycles) rather than spatial patterns (clusters). The EG concentration index serves as an external proxy for the sharpness of the scale-congestion trade-off in the present framework. Syverson (2004) provides direct evidence of the productivity-density relationship that underlies the hump-shaped profit assumption.

# 2 Environment

The environment is largely comprised of three elements. The first is a hump-shaped profit landscape that gives the industry a well-defined optimal scale. The second is a matching function for entry that endogenously generates a quadratic adjustment cost. The third is a linear-utility household, which lets industries evolve independently. Section 2.1 sets up the agents. Section 2.2 describes the profit landscape, Section 2.3 derives the timing friction, and Section 2.4 combines the ingredients into the reduced-form dynamics that drive the equilibrium.

## 2.1 Agents

The economy has two types of agents. A representative household consumes and supplies labor. A fixed mass of potential entrepreneurs alternates between operating firms and searching for entry opportunities. The household has linear utility over consumption, $U = \int_0^\infty e^{-\rho t} C_t \, \mathrm{d}t$, and supplies labor perfectly elastically at a fixed wage $\bar{w}$.[1] Three ingredients define the entry side of the environment.

**Assumption 1** (Entry economy)**.**
*A mass $M > 0$ of potential entrepreneurs, each at any instant either* active *(operating a firm) or* searching *(a potential entrant). Active firms produce using labor hired from the representative household at wage $\bar{w}$, with technology $y_i = z l_i^\gamma$, $\gamma \in (0, 1)$. Each entry incurs a per-firm sunk cost $\kappa \geq 0$, paid at the moment of entry. Firms exit at Poisson rate $\delta > 0$, returning to the search pool.*

[1] Equivalently, the household has linear utility in consumption and linear disutility from labor, $U = \int_0^\infty e^{-\rho t} \left[ C_t - \bar{w} \, L_t \right] \mathrm{d}t$, so the static labor-supply condition pins $w = \bar{w}$ and labor is supplied elastically. This is in the spirit of Greenwood et al. (1988) preferences in eliminating the wealth effect on labor supply.

Let $n_t$ denote the mass of active production units at date $t$ and $S_t = M - n_t$ the mass of searchers. Throughout the paper, I interpret $n$ as the mass of active *establishments* (physical production units such as plants, stores, or branches) rather than firms as legal entities. This interpretation matches both empirical series used in Sections 4–4.4. BEA investment in structures measures the rate of new establishment construction, and BDS establishment entry and exit rates measure establishment births and deaths. The matching-congestion microfoundation of $\mu$ (Section 2.3) is also most natural at the establishment level, where bottlenecks in permits, specialist labor, equipment, and sites constrain rapid adjustment. The aggregate state of the economy is fully characterized by $n_t$ and its rate of change $\dot{n}_t$.

## 2.2 Matching efficiency and the profit landscape

Each firm's effective output depends on market thickness. More firms create denser networks of suppliers, customers, and complementary services, improving efficiency. But beyond a point, additional firms congest the market. A scale efficiency function $m(n)$ multiplies each firm's output: $y_i = z l_i^\gamma \cdot m(n)$, with productivity $z$ and labor $l_i$. Profit is $\pi(n) = \bar{\pi}(\bar{w})\, m(n)$, where $\bar{\pi}(\bar{w})$ is the standard per-firm profit at the fixed wage $\bar{w}$ of Section 2.1, earned absent any scale effect.[2]

**Assumption 2** (Log-quadratic scale efficiency)**.**
*Scale efficiency takes the log-quadratic form:*

$$m(n) = \exp\left(-\frac{\varphi}{2}(n - n_p)^2\right), \qquad \varphi > 0,\ n_p > 0. \tag{2}$$

The function peaks at $n = n_p$ (optimal market scale) and falls off symmetrically. The parameter $\varphi$ governs the sharpness of the trade-off. High $\varphi$ means a narrow peak; low $\varphi$ means a broad peak. The hump-shaped specification is consistent with the productivity-density relationship documented in Syverson (2004), in which establishment-level productivity rises and then falls with market-level density. That is the empirical pattern encoded in the profit landscape. Online Appendix A.2 provides a matching-theoretic microfoundation.

For the analytical results, I work with the reduced-form log-quadratic profit:

$$\pi(n) = \pi_0 \exp\left(-\frac{\varphi}{2}(n - n_p)^2\right), \qquad \pi_0 > 0. \tag{3}$$

The constant $\pi_0 = \bar{\pi}(\bar{w})$ is the per-firm profit at the optimal scale $n_p$, where the scale efficiency attains its peak $m(n_p) = 1$; profit falls below $\pi_0$ as the industry's scale departs from $n_p$, at a

[2] Solving the firm's static problem $\max_l z l^\gamma - \bar{w} l$ gives optimal labor $l^* = (\gamma z/\bar{w})^{1/(1-\gamma)}$ and standard profit $\bar{\pi}(\bar{w}) = (1-\gamma)\gamma^{\gamma/(1-\gamma)} z^{1/(1-\gamma)} \bar{w}^{-\gamma/(1-\gamma)} > 0$.

rate set by the curvature $\varphi$. The semi-elasticity is exactly linear:

$$f(n) \equiv \frac{\pi'(n)}{\pi(n)} = -\varphi(n - n_p). \tag{4}$$

This linearity delivers exact harmonic oscillations (Theorem 1). The general case, where wage congestion adds a non-quadratic correction to $\ln \pi$, produces non-harmonic oscillations with the same leading-order frequency (Online Appendix B.3).

## 2.3 The timing friction: congestion in the entry process

Entry is mediated by a Cobb-Douglas matching technology $m(S, R) = S^{\alpha} R^{1-\alpha}$ with elasticity $\alpha \in (0, 1)$. The $S_t = M - n_t$ searchers compete for $R_t$ units of *entry resources* (permits, specialized labor, venture capital, factory sites) to form matches. The resource stock $R$ is endogenously supplied by an infrastructure sector. The matching-friction derivation in this subsection uses $R$ at its steady-state level $R_{ss}$ to pin the timing friction $\mu$. Deviations of $R$ from $R_{ss}$, driven by net adjustment pressure $\dot{n}$ in the stationary cycle, make the break-even entry cost $\kappa_t$ fluctuate, which pins cycle amplitude; Section 3.6 takes up that channel.

The matching infrastructure is tuned to clear the steady-state throughput $e_{ss} = \delta n_p$, with routine replacement at rate $\delta n_t$ serviced smoothly. The timing friction operates on net adjustment pressure $\dot{n}$, meaning deviations of the adjustment-speed margin from zero. Rapid entry ($\dot{n} > 0$) overloads permitting and construction, while rapid net contraction ($\dot{n} < 0$) overloads resale and reorganization. Both reduce the matching probability per searcher and produce the timing friction.

**Assumption 3** (Timing friction)**.**
*The multiplicative timing friction takes the log-quadratic form*

$$\xi(\dot{n}) = e^{-\frac{\mu}{2}\dot{n}^2}, \qquad \mu > 0,$$

*where $\mu$ is the friction parameter.*

The functional form arises as the continuous-time limit of *proportional adjustment costs* in the entry pipeline. In a discrete-time economy, suppose effective profit is reduced each period by a fraction proportional to the squared adjustment rate $(\varDelta n/\varDelta)^2$. Taking the period length $\varDelta \to 0$ yields the log-quadratic form, with $\mu$ a free friction parameter (Online Appendix A.3). The log-quadratic specification is canonical. Any symmetric friction $\xi$ with $\xi(0) = 1$ and $\xi'(0) = 0$ admits the second-order expansion $\ln \xi(\dot{n}) = -\frac{\mu}{2}\dot{n}^2 + O(\dot{n}^4)$. Symmetry of the flow penalty in $\dot{n}$ is a knife-edge condition. Under asymmetric strain, the leading-order ironing condition would carry a cubic term in $\dot{n}$ that produces amplitude-dependent frequency (Online Appendix B.5). The exact-harmonic result of Theorem 1 requires the symmetric specification, while the qualitative cycle survives asymmetric strain.

Combining Assumption 3 with the Cobb-Douglas matching technology pins the friction parameter to a specific function of structural primitives. Any symmetric strain term $g(\dot{n}^2/(\delta n_p)^2)$ with $g(0) = 0$ and $g'(0) = 1/2$, multiplying the matching probability, gives the following expression at second order around $\dot{n} = 0$:

$$\mu = \frac{1-\alpha}{\alpha\,(\delta n_p)^2}, \tag{5}$$

independent of the higher-order shape of $g$ (Online Appendix A.4). Here $\alpha$ is the matching elasticity and $\delta n_p$ is the steady-state replacement throughput. Equation (5) is the calibration I adopt when reading $\mu$ off the matching microfoundation; the body's results that depend on $\mu$ alone (e.g., $\omega = \sqrt{\varphi/\mu}$) hold under Assumption 3 for any positive friction parameter.

The timing friction $\mu$ decreases in the matching elasticity $\alpha$ and in the steady-state throughput $\delta n_p$. The entry-resource stock $R$ is absorbed into the steady-state calibration of $n_p$ through the matching technology, so it does not appear in (5) as a separate factor. Effective profit is $y_t = \pi(n_t)\,\xi(\dot{n}_t)$. The quadratic form in $\dot{n}$ arises generically from any smooth symmetric matching friction. Online Appendix A.3 derives the same $\xi$ from discrete-time proportional adjustment costs.

This functional form makes the friction depend on the rate of net adjustment $\dot{n}$ rather than on the level $n$ alone. That is the natural object when neighboring dates draw on the same permits, specialist construction crews, and equipment supply chains. The entry pipeline absorbs steady throughput at $\delta n_p$ smoothly, but it is strained when entries are packed into nearby dates. The relevant strain object is therefore local entry density along the time index, which is exactly $\dot{n}$. In a cross-sectionally indexed search model, neighboring submarkets are not physically linked, so there is no comparable strain object. "Rapid adjustment across the index" has no physical infrastructure to strain. The equilibrium dynamics developed below are built on this time-contiguity friction.

## 2.4 Combined structure

Combining the log-quadratic profit and the log-quadratic friction, effective profit is:

$$y_t = \pi_0\,\exp\!\Big(-\frac{\varphi}{2}(n_t - n_p)^2 - \frac{\mu}{2}\dot{n}_t^2\Big)\,. \tag{6}$$

The effective profit is a bivariate log-quadratic kernel in $(n, \dot{n})$, centered at $(n_p, 0)$. Both terms impose quadratic penalization of deviation from the optimum: $n_p$ for market scale and $\dot{n} = 0$ for adjustment speed.

# 3 The timing-search cycle

The equilibrium condition that agents cannot profit by retiming their entry, the ironing condition, pins a rate of entry along any approach to optimal scale. That pinned rate is one the industry cannot discard at the optimum without breaking equalization across neighboring submarkets along the time index. The result is cycles, and in the deterministic case they are exactly harmonic with frequency $\omega = \sqrt{\varphi/\mu}$. This section formalizes the argument.

## 3.1 The agent's problem

A searching agent chooses a date $\tau$ at which to enter. Upon entry, she operates a firm earning the effective profit $y_t = \pi(n_t)\,\xi(\dot{n}_t)$ of Section 2.4 per unit time, discounted at rate $r \geq 0$, with exogenous exit at rate $\delta > 0$. The value of entering at date $\tau$ is:

$$V(\tau) = \int_{\tau}^{\infty} e^{-(r+\delta)(t-\tau)}\, y_t \,\mathrm{d}t. \tag{7}$$

$V(\tau)$ is the value of a single entry spell: operating from $\tau$ until stochastic exit. Re-entry after exogenous exit is treated as a fresh draw of the same problem for whichever agent occupies the search pool at that date. Under the ironing condition, the single-spell formulation is therefore without loss for equilibrium dynamics. The agent maximizes $V(\tau)$ over $\tau$, taking the aggregate path $\{n_t\}$ as given. An agent who does not enter at date $t$ remains in the search pool, with continuation value

$$V_t^{\text{search}} = \max_{\tau \geq t} e^{-r(\tau - t)} \big[V(\tau) - \kappa\big], \tag{8}$$

where $\kappa$ is the per-firm sunk entry cost from Assumption 1, paid at the moment of entry.

## 3.2 Equilibrium

An *entry policy* is a measurable function $e : \mathbb{R}_+ \to \mathbb{R}_+$ specifying the flow rate of entrants at each date. Together with the initial condition $n_0$, it determines the entire path of the economy through the aggregate consistency condition $\dot{n}_t = e_t - \delta n_t$.

**Definition 1** (Timing search equilibrium)**.**
*A* timing search equilibrium *is a path* $\{n_t\}_{t\geq 0}$ *and an entry policy* $e(\cdot)$ *such that:*

*(i)* ***Individual optimality****: Each agent maximizes her discounted continuation value, comparing immediate entry against remaining in the search pool and entering later, with search value* $V_t^{search}$ *of* (8)*.*

*(ii)* ***Full-support free entry (ironing)****: entry occurs at every date, and unrestricted entry into the search pool (*$V_t^{search} = 0$*) equalizes the value of entry to its cost,* $V(\tau) = \kappa$ *for every* $\tau$*.*

*(iii)* ***Aggregate consistency****:* $\dot{n}_t = e_t - \delta\, n_t$*, where* $e_t \geq 0$*.*

*(iv)* ***Feasibility****:* $n_t \leq M$ *for all* $t$*.*

**Free entry, strengthened by full support** Free entry into the search pool ($V_t^{\text{search}} = 0$) is maintained throughout. Ordinary free-entry complementarity requires only $V_t = \kappa$ at active dates and $V_t \leq \kappa$ where no entry occurs, which permits inactive dates and admits other equilibria. Condition (ii) strengthens this by selecting equilibria with *full temporal support*: entry at every date, so $V(\tau) = \kappa$ everywhere. This is the key requirement, the temporal analog of the equal-profit condition in Burdett and Mortensen (1998), where no firm gains by reposting its wage; here no agent gains by retiming her entry, so if any date offered strictly higher value, agents would crowd it, changing $n_t$ until the advantage vanished. Its differential form (Proposition 1) generates the cycle.

**Commitment versus dynamic optimization** The formulation in (7)–(8) is a commitment problem: agents choose $\tau$ taking the entire future path $\{n_t\}$ as given. An alternative formulation would have agents observe the current state $(n_t, \dot{n}_t)$ at each instant and decide dynamically whether to enter. The ironing condition makes the two equivalent. Since $V(\tau) = V^*$ for all $\tau$ in equilibrium, an agent who could re-optimize at any date $t$ would find no profitable deviation. She is indifferent across all entry dates. The commitment equilibrium is therefore also an equilibrium of the dynamic problem: agents who can freely re-time their entry choose not to.

## 3.3 The ironing condition

The key equilibrium condition is that agents cannot profit by retiming their entry. This is the temporal analog of the equal-profit condition in Burdett and Mortensen (1998). Burdett-Mortensen equalize profits across wages so that no firm gains by reposting. Timing search equalizes payoffs across dates so that no agent gains by retiming: this is the full-support temporal indifference of Definition 1(ii), which I call *ironing*. Proposition 1 translates it into the differential condition on the equilibrium path that generates the cycle.

**Proposition 1** (Differential ironing condition)**.**
*Full-support temporal indifference (Definition 1(ii)), $V(\tau) = \kappa$ for all $\tau$, implies the differential ironing condition equalizing the effective flow payoff at the break-even level:*

$$\pi(n_t)\,\xi(\dot{n}_t) = (r+\delta)\kappa \qquad \textit{for all } t. \tag{9}$$

*Proof.* By full-support temporal indifference (Definition 1(ii)), entry occurs at every date and $V(\tau) = \kappa$ for all $\tau$. The equilibrium path $n(\cdot)$ is $C^1$ by aggregate consistency (condition (iii)) and the smoothness of $\pi$ and $\xi$, so $V(\tau)$ is differentiable in $\tau$.[3] Differentiating (7):

$$\frac{\mathrm{d}V}{\mathrm{d}\tau} = -\pi(n(\tau))\,\xi(\dot{n}(\tau)) + (r+\delta)\,V(\tau).$$

Setting $\frac{\mathrm{d}V}{\mathrm{d}\tau} = 0$ (since $V \equiv \kappa$): $\pi(n(\tau))\xi(\dot{n}(\tau)) = (r+\delta)\kappa$. ∎

Proposition 1 is the ironing condition of the deterministic model of this section (constant entry cost $\kappa_t = \bar{\kappa}$), where it holds pathwise. Under shocks the value of entry fluctuates with the state, so the pathwise statement no longer holds; it generalizes to the stochastic ironing condition (15) of Section 3.6 (the martingale form of Definition 2(ii)), which adjusts the break-even annuity for the expected capital gain in firm value and recovers (9) as $\sigma_\kappa \downarrow 0$.

**Free entry and the break-even payoff** With free entry into the search pool, the value of entry is competed to the entry cost at every active date, $V_t = \kappa$, so no entry date carries a positive surplus: a firm started at any date just recovers $\kappa$. The ironing condition (9) is the differential form of this break-even condition; it fixes the effective flow payoff at $(r+\delta)\kappa$, the level at which discounted operating profit exactly covers the entry cost. The cross-date equalization is the temporal analog of the cross-wage equal-profit condition in Burdett and Mortensen (1998), with the equalized value tied down here by the entry cost rather than left as a free constant.

**Regularity condition** The condition that entry occurs at all dates is satisfied whenever the searcher pool $S_t = M - n_t$ is bounded away from zero along the equilibrium path and

[3] The differentiability of $V$ in $\tau$ requires that $t \mapsto \pi(n_t)\xi(\dot{n}_t)$ is locally integrable along the equilibrium path, which holds whenever $n(\cdot) \in C^1$. This regularity is guaranteed by the smoothness of $\pi$ (Assumption 2) and $\xi$ (Section 2.3), combined with the continuity of the entry flow $e(\cdot)$ in condition (iii).

the timing friction $\mu$ is positive. With a large population $M$ and moderate cycle amplitude $A < M - n_p$, the condition holds. Online Appendix B.4 provides a formal verification.

### 3.4 The deterministic cycle

The equilibrium under log-quadratic profits and a constant entry cost is an exact harmonic oscillator with frequency $\omega = \sqrt{\varphi/\mu}$ (Theorem 1). This subsection derives that result. The constant-entry-cost case is not the most realistic specification, but it is the most analytically tractable one. The ironing condition reduces to a harmonic oscillator whose path and period admit closed-form expressions at every amplitude.[4] Taking logarithms of the ironing condition (9), substituting the functional forms (6), and rearranging:

$$\underbrace{\frac{\varphi}{2}(n_t - n_p)^2}_{\text{profit loss}} + \underbrace{\frac{\mu}{2}\,\dot{n}_t^2}_{\text{adjustment cost}} = \underbrace{b}_{\substack{\text{profit shortfall}\\ \text{(pinned by }\kappa)}}, \tag{10}$$

where $b \equiv \ln \dfrac{\pi_0}{(r+\delta)\kappa} > 0$ is the *per-firm profit shortfall*, equivalently the *break-even profit gap*: the log gap between the optimal-scale peak profit $\pi_0$ and the break-even flow payoff $(r+\delta)\kappa$ that ironing holds the industry's effective profit to. Free entry holds it constant along the cycle and pins it to the entry cost $\kappa$. Differentiating Equation (10) with respect to time gives $\dot{n}_t\,[\varphi(n_t - n_p) + \mu\ddot{n}_t] = 0$. On intervals where $\dot{n}_t \neq 0$, dividing through by $\dot{n}_t$ yields $\varphi(n_t - n_p) + \mu\ddot{n}_t = 0$. At the two turning points where $\dot{n}_t = 0$, the same identity holds by continuity, since the bracket is a smooth function of $(n, \ddot{n})$ that vanishes on a dense set. The resulting law of motion is

$$\underbrace{\ddot{n}_t}_{\substack{\text{change in}\\ \text{entry rate}}} = \underbrace{-\frac{\varphi}{\mu}\,(n_t - n_p)}_{\substack{\text{pull toward}\\ \text{optimal scale}}}. \tag{11}$$

At every instant the change in the entry rate is pinned by the current deviation from optimal scale, scaled by the ratio of profit curvature to timing friction $\varphi/\mu$: industries far from $n_p$ accelerate quickly back toward it, industries near $n_p$ adjust slowly. This is the deterministic skeleton of the timing-search dynamics. Section 3.6 adds fluctuating entry costs to the same skeleton, but the curvature-to-friction ratio $\varphi/\mu$ that pins the structural frequency in Theorem 1 is unchanged through both extensions.

[4] Section 3.6 extends this oscillator with fluctuating entry costs; the same closed-form frequency $\omega = \sqrt{\varphi/\mu}$ carries over exactly to the stationary cycle of that section, where it locates the spectral peak of $n_t$ (Proposition 4).

**Theorem 1** (Exact harmonic oscillations)**.**
*In a timing search equilibrium (Definition 1) under Assumptions 1–3, with entry at all dates ($A \leq \delta n_p/\sqrt{\omega^2+\delta^2}$; verified below), the equilibrium path is exactly*

$$n_t = n_p + A\cos(\omega t+\phi), \qquad \omega = \sqrt{\frac{\varphi}{\mu}}, \qquad A = \sqrt{2b/\varphi}. \tag{12}$$

*Proof.* The ironing condition (Proposition 1) gives (10). Differentiation yields the harmonic oscillator equation (11), whose general solution is (12) with $\omega = \sqrt{\varphi/\mu}$. The solution is exact because the log-quadratic specification makes the semi-elasticity $f(n)$ exactly linear (equation (4)), so no Taylor approximation is needed. Substituting back: $\frac{\varphi}{2}A^2\cos^2(\omega t+\phi) + \frac{\mu}{2}\omega^2 A^2 \sin^2(\omega t+\phi) = \frac{\varphi}{2}A^2 = b$. ∎

The cycle, both its period and its size, is neutral to the level of productivity $z$. Productivity enters profit multiplicatively, $\pi(n) = \bar{\pi}(\bar{w})\, m(n)$ with $\bar{\pi}(\bar{w}) \propto z^{1/(1-\gamma)}$, so a change in $z$ rescales the whole profit curve by a common factor. The dynamics, however, run on the profit shortfall $b = \ln(\pi_0/y_t)$, the log gap between current effective profit $y_t = \pi(n_t)\xi(\dot{n}_t)$ and its peak $\pi_0 = \pi(n_p)\xi(0)$. A common rescaling multiplies $y$ and $\pi_0$ alike and cancels in the ratio, so $b = \frac{\varphi}{2}(n_t - n_p)^2 + \frac{\mu}{2}\dot{n}_t^2$ contains no $z$. The frequency $\omega = \sqrt{\varphi/\mu}$ is set by the curvature of $\ln \pi$, equivalently of the scale efficiency $\ln m$, which a level shift leaves untouched; and the amplitude $A = \sqrt{2b/\varphi}$ inherits this $z$-independence from $b$, as does the stochastic amplitude of Section 3.6. What $z$ moves is the level on which the entry cycle rides: profit and output per firm scale with $z$, and with them the absolute magnitude of the cycle, while the cycle in the number of firms, its period and proportional size alike, does not. Productivity raises the reward for operating uniformly across scales, shifting neither the optimal scale $n_p$ nor the steepness $\varphi$ of the profit landscape that together generate the cycle.

The pair $(n_t, e_t)$ is a timing search equilibrium for any amplitude $A \leq \delta n_p/\sqrt{\omega^2+\delta^2}$, the binding sufficient condition for $e_t = \dot{n} + \delta n \geq 0$ throughout the cycle (Online Appendix B.6). Industries with net entry routinely $|\dot{n}| > \delta n_p/2$ are outside the regime where the ironing condition holds globally. The amplitude is pinned by the entry-cost level in the stochastic equilibrium of Theorem 2, a structural quantity recovered from the data.

**Structural decomposition** The frequency $\omega = \sqrt{\varphi/\mu}$ decomposes into two primitives. The numerator $\varphi$ is the curvature of the profit landscape: how sharply profits fall when the industry deviates from optimal scale. The denominator $\mu$ is the timing friction: how costly it is to adjust the number of active firms rapidly, arising from congestion in the entry matching process. Treated as an abstract pair, $(\varphi, \mu)$ alone determines $\omega$; the frequency does not depend on the realized exit rate $\delta$ or on the shock process (Section 3.6). When $\mu$ is calibrated structurally via Assumption 3, $\mu = (1-\alpha)/[\alpha(\delta n_p)^2]$ inherits dependence on the matching elasticity $\alpha$ and the steady-state throughput $\delta n_p$; the resource-stock $R$ underlying the matching technology is absorbed into the steady-state calibration of $n_p$ and does not appear separately in the formula. The structural decomposition is therefore: $\omega$ depends on $(\varphi, \mu)$ as primitives, and on $(\alpha, \delta, n_p)$ when $\mu$ is read off the microfoundation. Substituting $\mu = (1-\alpha)/[\alpha(\delta n_p)^2]$ into $\omega = \sqrt{\varphi/\mu}$ gives the structural frequency

$$\omega = \delta n_p \sqrt{\alpha\varphi/(1-\alpha)}, \tag{13}$$

connecting cycle frequency to three independently measurable objects: matching elasticity $\alpha$, profit curvature $\varphi$, and steady-state turnover $\delta n_p$. In the absence of direct estimates of entry-matching elasticities, I use the labor-market range $[0.5, 0.7]$ from Petrongolo and Pissarides (2001) as a benchmark; the structural frequency is robust to $\alpha \in [0.3, 0.8]$ since $\mu$ varies smoothly with $\alpha$ over this range. Table I contrasts this decomposition with the way standard frameworks pin cycle frequency.

**Multi-industry general equilibrium** Because the wage is fixed and households have linear utility, each industry's profit landscape depends only on its own number of active firms. Under idiosyncratic shocks and no inter-industry input-output linkages or demand spillovers, a multi-industry economy with $I$ industries each characterized by $(\varphi_i, \mu_i)$ evolves as $I$ independent timing-search systems, and the aggregate spectral density is a sum of industry-specific peaks at frequencies $\omega_i = \sqrt{\varphi_i/\mu_i}$.[5] The isolation of frequency from the shock process is therefore a *cross-sectional* prediction: different industries cycle at different frequencies determined by their respective market structures. Under common shocks or input-output linkages, this additive spectral structure does not survive aggregation. Section 6 bounds the scope of the closed-form result, with quantitative leakage of $(\varphi, \mu)$ into the welfare cost under CRRA preferences documented in Online Appendix E.1.

[5] This sum is a microfounded Fourier decomposition of the aggregate cycle. The ironing condition pins each unit's optimal entry path as the harmonic function $n_{i,t} = n_{p,i} + A_i \cos(\omega_i t + \phi_i)$ at its own structural frequency $\omega_i = \sqrt{\varphi_i/\mu_i}$: the individual optimal decision is itself a Fourier basis element by construction, not a basis element imposed from outside. Pushed to a continuous heterogeneity index, the aggregate cycle $X_t = \int n_{i,t}\, dF(i)$ is the weighted sum of these basis elements under the heterogeneity distribution $F$ over $(\varphi, \mu)$, with bandwidth set by the entry-cost persistence $\theta$ (Section 3.6). Because the timing friction $\mu$ sets frequency but not amplitude (Theorem 3), cross-industry dispersion in $\mu$ spreads a conserved aggregate variance across frequencies without changing it: a homogeneous economy shows a sharp aggregate cycle and a heterogeneous one shows the same volatility as broadband fluctuation, so aggregate cyclicality reflects market-structure homogeneity rather than shock size. The decomposition relies on three conditions: idiosyncratic shocks, no input-output linkages, and linear utility. With granular establishment- or region-level data, the empirical aggregate spectrum can be read as a structural deconvolution problem on $F$, an extension that current 3-digit cross-sectional tests do not exploit.

Table I: What determines cycle frequency?

| | **Frequency determined by** | **Cross-industry prediction** |
|---|---|---|
| Canonical RBC | Shock autocorrelation | Common if shocks are common |
| Canonical NK | Policy rule | Common across industries |
| Beaudry et al. (2020) | Bifurcation condition | Not the object of their analysis |
| **Timing search** | $\boldsymbol{\omega = \sqrt{\varphi/\mu}}$ | **Industry-specific frequency** |

*Notes:* In canonical RBC and NK specifications, the spectral peak reflects the shock process or the policy rule, objects common across industries; multi-sector enrichments can produce dispersion but do not, by themselves, impose a $\gamma$-dependent slope (Online Appendix F.2). In Beaudry et al. (2020), the frequency emerges from a Hopf bifurcation; cross-industry frequency variation is not the object of their analysis. In timing search, the frequency is a closed-form function of two market-structure primitives ($\varphi$: profit-landscape curvature; $\mu$: timing friction), is exact under linear utility, and is isolated from the shock process (Proposition 3).

**An analogy: a spring with stiffness $\varphi$ and inertia $\mu$** As an industry expands toward optimal scale, firms keep entering because profits are above the steady-state norm. In directed-search vocabulary, this expansion is an adjustment across temporally adjacent submarkets, with each date a submarket of its own. The ironing condition, which says that no firm can profit by entering at a different date, requires the value of entry to be equalized across these adjacent submarkets. Because neighboring dates are physically contiguous through the entry pipeline, this equalization translates into a smooth entry rate over time: if the rate jumped or changed abruptly, firms entering just before the jump would face matching conditions different from firms entering just after, and the equalization would break. As the industry approaches $n_p$, the profit landscape flattens because $\pi'(n)$ approaches zero, and the level benefit of further entry saturates. Yet the entry flow built up across the prior cross-submarket adjustment is positive at $n_p$ and cannot drop discontinuously to zero there without breaking indifference. The momentum from the prior adjustment therefore carries the entry flow through $n_p$, and the industry overshoots optimal scale. Beyond $n_p$, the same indifference logic drives entry to the other side of replacement, the flow reverses, and the industry oscillates rather than settling. The economics maps cleanly onto classical mechanics. Profit curvature $\varphi$ acts as a *restoring force* (steeper fall, harder pull back), and the timing friction $\mu$ acts as *inertia* (larger $\mu$, harder to change the pace of entry). A mass-spring system with spring stiffness $k$ and mass $m$ oscillates at $\omega = \sqrt{k/m}$ for exactly the same reason: stiffness pulls the mass back toward equilibrium, and mass resists changes in velocity. Here profit curvature $\varphi$ plays the role of stiffness $k$ and timing friction $\mu$ plays the role of mass $m$, so the timing-search industry oscillates at $\omega = \sqrt{\varphi/\mu}$. Both forces are necessary. Without the scale externality the spring has no stiffness; without the matching friction the industry has no inertia, and the system returns to the optimum without oscillating.

The same spring logic explains why the period does not depend on how hard the industry is hit. A deep recession leaves the industry far below optimal scale, profits per active firm are well above norm, and the ironing condition sustains a fast inflow; but the deeper trough means the industry has further to travel. A small shock generates weaker incentive, slower entry, and a shorter path. Speed and distance scale together, so $T = 2\pi\sqrt{\mu/\varphi}$ is the same in either case. This depth-independence extends to the stochastic equilibrium: successive swings preserve their period even as their amplitude decays (Online Appendix C.1). A direct corollary is the plucking pattern: deeper recessions recover faster (peak recovery speed $A\omega$ scales linearly with amplitude) but in the same time, the timing-search analogue of the plucking pattern documented by Friedman (1993), Bordo and Haubrich (2017), and Dupraz et al. (2019), and tested in Section 4.4.

The closed form has several immediate consequences for the cycle, collected next.

**Corollary 1** (Properties of the deterministic cycle)**.** *Under Theorem 1:*

*(i) the frequency rises with profit curvature and falls with the timing friction, $\partial\omega/\partial\varphi > 0$ and $\partial\omega/\partial\mu < 0$;*

*(ii) the period $T = 2\pi\sqrt{\mu/\varphi}$ depends only on the market-structure pair $(\varphi, \mu)$;*

*(iii) the frequency is the same whether the cycle is deep or shallow, and is unaffected by productivity $z$, which scales profits without changing their curvature.*

*Proof.* Items (i)–(iii) follow by direct calculation from $\omega = \sqrt{\varphi/\mu}$ in Theorem 1, which contains no dependence on $(z, A)$. ∎

**Beyond the log-quadratic** The closed-form result extends to general hump-shaped profits: under any $C^2$ profit function $\pi$ with $\pi'(n_p) = 0$ and $\pi''(n_p) < 0$, the frequency takes the form $\omega = \sqrt{|\pi''(n_p)|/[\mu\,\pi(n_p)]} + O(A^2)$, and the log-quadratic-based prediction $T_0 = 2\pi\sqrt{\mu/\varphi}$ stays within 1.6% of the exact full-cycle period at $A/n_p = 0.30$ across 8 smooth hump-shaped profit specifications (Online Appendix B.3 and C.2). The log-quadratic profile is the special case in which the finite-amplitude frequency correction vanishes, so the period is exactly amplitude-independent (Online Appendix B.5). A skewed profit landscape produces the empirically standard asymmetry in expansions and contractions: when profit falls faster on the congestion side, the economy spends more time below $n_p$, giving slow expansions and sharp contractions (Online Appendix B.5).

**Contrast with bifurcation-driven cycles** The stochastic cycle of Theorem 2 is a unique stationary object: under a fluctuating entry cost, the amplitude distribution is drawn exponentially fast (at rate $\theta$) to the same stationary law at $\sqrt{b/\varphi}$, with the phase uniform, the unique rotation-invariant law. This route differs in kind from the bifurcation-driven route familiar from Beaudry et al. (2020), where a supercritical Hopf bifurcation produces an isolated, asymptotically stable limit cycle whose amplitude is a deterministic feature surviving small parameter perturbations. Here the period is fixed by $(\varphi, \mu)$ while the amplitude is pinned by the entry-cost level $\bar{\kappa}$, a structural quantity recovered from the data. Both routes deliver stable cycles, but the amplitude is pinned by different primitives: the bifurcation condition in the Hopf case, the entry-cost level here.

## 3.5 Balanced growth cycle

The closed-form result is stated around a stationary $n_p$, but the model accommodates balanced growth at the level of primitives provided three homogeneity conditions hold: intensive-form profit $\pi(n,t) = z_t\hat{\pi}(n/n_{p,t})$ with $z_t$ and $n_{p,t}$ trending at exogenous rate $g$ and $\hat{\pi}$ hump-shaped around 1; matching technology $m(S,R) = S^{\alpha}R^{1-\alpha}$ homogeneous of degree 1 (already satisfied); and timing friction in percentage rates, $\xi(\dot{n}/n) = \exp(-\frac{\mu}{2}(\dot{n}/n)^2)$.[6] The household discount rate satisfies $\rho > g$.

**Proposition 2** (Balanced growth cycle, small-amplitude)**.**
*Under the homogeneity setup above, the equilibrium of Theorem 1 extends to balanced growth: the detrended log deviation $\tilde{x}_t = \ln n_t - \ln n_{p,t}$ satisfies, to leading order,*

$$\mu\,\ddot{\tilde{x}} + \varphi\,\tilde{x} = 0, \qquad \tilde{x}_t = \tilde{A}\cos(\omega t + \phi), \qquad \omega = \sqrt{\varphi/\mu}, \tag{14}$$

*with $\varphi \equiv -\hat{\pi}''(1)/\hat{\pi}(1)$ the intensive-form log-curvature. Higher-order terms give $O(\tilde{A}^2)$ corrections.*

*Proof.* Detrending the agent's value function with effective discount $\rho - g > 0$ yields a stationary problem in $\tilde{n}$. The ironing condition becomes $\hat{\pi}(\tilde{n})\,\xi(\dot{\tilde{n}}/\tilde{n}) = \tilde{C}$. Linearization around $\tilde{n} = 1$ delivers (14). ∎

**The BGP amplitude** As written, equation (14) is the deterministic BGP cycle. With a fluctuating entry cost, the detrended profit shortfall $\tilde{b} = \frac{\varphi}{2}\tilde{x}^2 + \frac{\mu}{2}\dot{\tilde{x}}^2$ mean-reverts at rate $\theta$ exactly as the profit shortfall does at a stationary optimum (Section 3.6), to the same

[6] The body's $\xi(\dot{n})$ coincides with $\xi(\dot{n}/n)$ at leading order around the stationary $n_p$; the percentage-rate version is the BGP-compatible primitive specification.

stationary level $\bar{b} = \bar{b}$. By Theorem 2, applied to the detrended variable, the cycle amplitude is pinned at $\sigma_{\tilde{x}} = \sqrt{\bar{b}/\varphi}$, now measured in detrended log-deviation units. The frequency $\omega = \sqrt{\varphi/\mu}$ and the amplitude $\sigma_{\tilde{x}}$ therefore depend on disjoint primitives on the balanced-growth path exactly as they do at a stationary optimum, so the frequency-welfare separation of Theorem 3 carries over to growing industries without modification.

Two implications. The frequency $\omega = \sqrt{\varphi/\mu}$ is invariant to the BGP rate $g$, so the cross-industry test in Section 4.3 extends to industries on different growth trends; and growth-side ($g$) and cycle-side ($\varphi, \mu, \bar{\kappa}$) policy interventions are structurally separable in the model, complementing the entry-side taxonomy of Section 5.3 with a growth-side dimension.

## 3.6 The stochastic entry cost and the stationary cycle

In the deterministic model free entry holds the entry cost fixed at $\bar{\kappa}$, so the profit shortfall is the constant $\bar{b} = \ln \frac{\pi_0}{(r+\delta)\bar{\kappa}}$ and the industry rides one cycle (Theorem 1). What makes the cycle stochastic is that entry costs fluctuate: permits, construction, equipment, and financing vary over time, so the break-even entry cost $\kappa_t$ mean-reverts around $\bar{\kappa}$. As $\kappa_t$ moves, free entry shifts the cycle the industry rides, and the cycle's amplitude varies with the entry cost. The rest of this section makes this precise.

**The stochastic entry cost** The resources to start a firm, permits, specialist crews, equipment, sites, and financing, are not constant: their cost fluctuates and is persistent. I model the break-even entry cost as a mean-reverting process.

**Assumption 4** (Stochastic entry cost)**.**
*The log entry cost follows an Ornstein–Uhlenbeck process,*

$$d \ln \kappa_t = -\theta \left(\ln \kappa_t - \ln \bar{\kappa}\right) dt + \sigma_\kappa \, dW_t,$$

*with long-run level $\bar{\kappa}$, persistence $\theta > 0$, and volatility $\sigma_\kappa > 0$.*

Free entry holds the value of an active firm at the entry cost, $V_t = \kappa_t$, and that value obeys the asset-pricing equation $(r + \delta)V_t = \pi(n_t)\xi(\dot{n}_t) + \mathbb{E}_t[dV_t]/dt$, so

$$\pi(n_t)\, \xi(\dot{n}_t) = (r + \delta)\, \kappa_t - \mathbb{E}_t[d\kappa_t]/dt. \tag{15}$$

This is the stochastic counterpart of ironing: across entry dates it equalizes *total expected returns*, current profit plus the expected capital gain $\mathbb{E}_t[d\kappa_t]/dt$ in firm value, where deterministic ironing ($d\kappa = 0$, Proposition 1) equalizes current profit alone. The break-even annuity is *adjusted* for that expected capital gain, not replaced by an expected entry cost. For the entry cost of Assumption 4 the adjustment is of the order of the fluctuation and slow, so $\pi(n_t)\xi(\dot{n}_t) \approx (r+\delta)\kappa_t$ and the profit shortfall tracks the entry cost, $b_t \approx \ln \frac{\pi_0}{(r+\delta)\kappa_t}$ (the exact Itô adjustment is computed in Online Appendix B.1 and absorbed into the calibrated $(\bar{b}, \sigma_\kappa)$). The level $\bar{\kappa}$ sets the mean profit shortfall $\bar{b} = \ln \frac{\pi_0}{(r+\delta)\bar{\kappa}}$ and hence the typical cycle amplitude; the persistence $\theta$ and volatility $\sigma_\kappa$ govern how the amplitude fluctuates around it. The matching friction $\mu$ plays no part here: it sets the frequency $\omega_0 = \sqrt{\varphi/\mu}$ but not the profit shortfall. This is the source of the frequency-welfare separation of Theorem 3: the frequency lives in the landscape pair $(\varphi, \mu)$, while the profit shortfall, and with it the welfare cost, lives in the disjoint entry-cost level $\bar{\kappa}$.

**Scope** The stochastic statements below are a small-fluctuation reading, leading order in the entry-cost fluctuation, and rest on two further commitments stated precisely in Online Appendix B.1: the entry cost varies slowly and smoothly relative to the cycle, so the net entry rate $\dot{n}_t = dn_t/dt$ stays well defined; and its response to net adjustment pressure (Section 2.3) holds the cycle away from zero and pins down a unique long-run distribution.

**The echo of an entry-cost shock** A one-time displacement of the entry cost shifts the profit shortfall, which then mean-reverts. This already delivers a distinctive cycle property the data can verify: absent further shocks, a deviation of $b_t$ from $\bar{b}$ decays at the persistence rate $\theta$,

$$b_t - \bar{b} \;=\; (b_0 - \bar{b})\, e^{-\theta t}, \tag{16}$$

so a displaced industry has its amplitude $A_t = \sqrt{2b_t/\varphi}$ revert toward $\sqrt{2\bar{b}/\varphi}$ at rate $\theta/2$ while it keeps cycling at the structural frequency $\omega_0 = \sqrt{\varphi/\mu}$. Three properties follow.

- *The period does not change.* Because the entry-cost shock acts on the profit shortfall rather than on the speed of adjustment, the cycle length stays exactly at the structural $T = 2\pi/\omega_0$ throughout the return.
- *The amplitude reverts toward its long-run level.* Each successive swing returns toward $\sqrt{2\bar{b}/\varphi}$, at a rate set by the persistence $\theta$; the amplitude does not diverge.
- *Period and amplitude reversion are independent.* Faster mean-reversion returns the amplitude faster but leaves the cycle length untouched, in contrast to AR(1) and similar linear-decay models where persistence and period are mechanically linked.

This is the *plucking / echo pattern*: a one-time shock triggers oscillations that lose amplitude over time while keeping their period. U.S. manufacturing investment after the Great Depression onset oscillates at $T = 5.0$ years through the early 1970s while the amplitude decays (Section 4.4); oil and mining investment after 1979 oscillates at $T = 3.25$ years. The pattern distinguishes the entry-cost mechanism from linear-decay alternatives such as AR(1), where decay rate and period are mechanically linked (Online Appendix C.1 plots the impulse responses).

**The fluctuating cycle** With $\kappa_t$ stochastic, the ironing condition holds in conditional expectation rather than pathwise: the profit shortfall $b_t = \ln \frac{\pi_0}{(r+\delta)\kappa_t}$ moves with the entry cost, and the industry shifts between cycles of different size. Being affine in $-\ln \kappa_t$, it inherits the entry cost's Ornstein–Uhlenbeck law,

$$db_t = -\theta\,(b_t - \bar{b})\,dt - \sigma_\kappa\,dW_t, \tag{17}$$

mean-reverting to $\bar{b}$ at rate $\theta$. The spectral peak of $n_t$ remains at the structural frequency $\omega_0 = \sqrt{\varphi/\mu}$.

**Proposition 3** (Stochastic limit cycle)**.**
*Under the stochastic entry cost (17):*

*(i) The spectral density of $n_t$ peaks at $\omega_0 = \sqrt{\varphi/\mu}$.*

*(ii) The peak depends only on $(\varphi, \mu)$, not on the entry-cost process $(\bar{\kappa}, \theta, \sigma_\kappa)$.*

*(iii) Entry-cost fluctuations produce persistent cycles.*

*Proof.* See Online Appendix B.1. ■

The structural frequency is a property of market structure, not the entry-cost process: entry-cost fluctuations produce a spectral peak at $\omega_0 = \sqrt{\varphi/\mu}$, with persistence shifting it negligibly when the frequency response is sharp. This matches the central finding of Beaudry et al. (2020), persistent boom-bust dynamics from short-lived disturbances, though the mechanism here is the scale-congestion trade-off rather than financial-friction complementarities; numerical confirmation is in Online Appendix C.1.

**The selected amplitude** Because $b_t = \ln \frac{\pi_0}{(r+\delta)\kappa_t}$ is affine in $-\ln \kappa_t$, it inherits the Ornstein–Uhlenbeck law of the entry cost and has a Gaussian stationary distribution,

$$b_t \sim \mathcal{N}\Big(\bar{b},\ \tfrac{\sigma_\kappa^2}{2\theta}\Big), \qquad \bar{b} = \ln \frac{\pi_0}{(r+\delta)\bar{\kappa}},$$

so the profit shortfall fluctuates around the level $\bar{b}$ set by the mean entry cost. Its mean $\bar{b}$ is the per-firm welfare cost of the cycle (Section 5.1). Because the entry cost acts on the profit shortfall, not on the profit curvature or timing friction that set the pace, it moves the cycle's amplitude, and hence its welfare cost, but not its frequency, which stays at $\omega_0 = \sqrt{\varphi/\mu}$.

**Proposition 4** (Stationary amplitude)**.**
*Under the stochastic entry cost of Assumption 4:*

*(i) The profit shortfall is stationary Gaussian,*

$$b_t \sim \mathcal{N}\Big(\bar{b},\ \tfrac{\sigma_\kappa^2}{2\theta}\Big), \qquad \bar{b} = \ln \frac{\pi_0}{(r+\delta)\bar{\kappa}}. \tag{18}$$

*(ii) The mean cycle amplitude $\sqrt{2\bar{b}/\varphi}$ is set by the entry-cost level $\bar{\kappa}$ and is independent of $\mu$.*

*(iii) The spectral peak sits at $\omega_0 = \sqrt{\varphi/\mu}$.*

*Proof.* See Online Appendix B.2. ■

Figure 1 visualizes the cycle's stochastic structure: after a one-time entry-cost shock the amplitude reverts toward its long-run level (top right), larger entry-cost swings spread the amplitude more widely (bottom left), and mean-reverting entry costs maintain a stationary cycle (bottom right) with $\mathbb{E}[(n_t - n_p)^2] = \bar{b}/\varphi$. Online Appendix C.1 confirms the stationary amplitude distribution numerically.

The mean amplitude $\sqrt{2\bar{b}/\varphi}$ does not depend on $\mu$: industries with a low entry-cost level $\bar{\kappa}$ or a flat profit landscape experience larger cycles, while $\mu$ governs only the cycle frequency. The per-firm welfare cost is the mean profit shortfall,

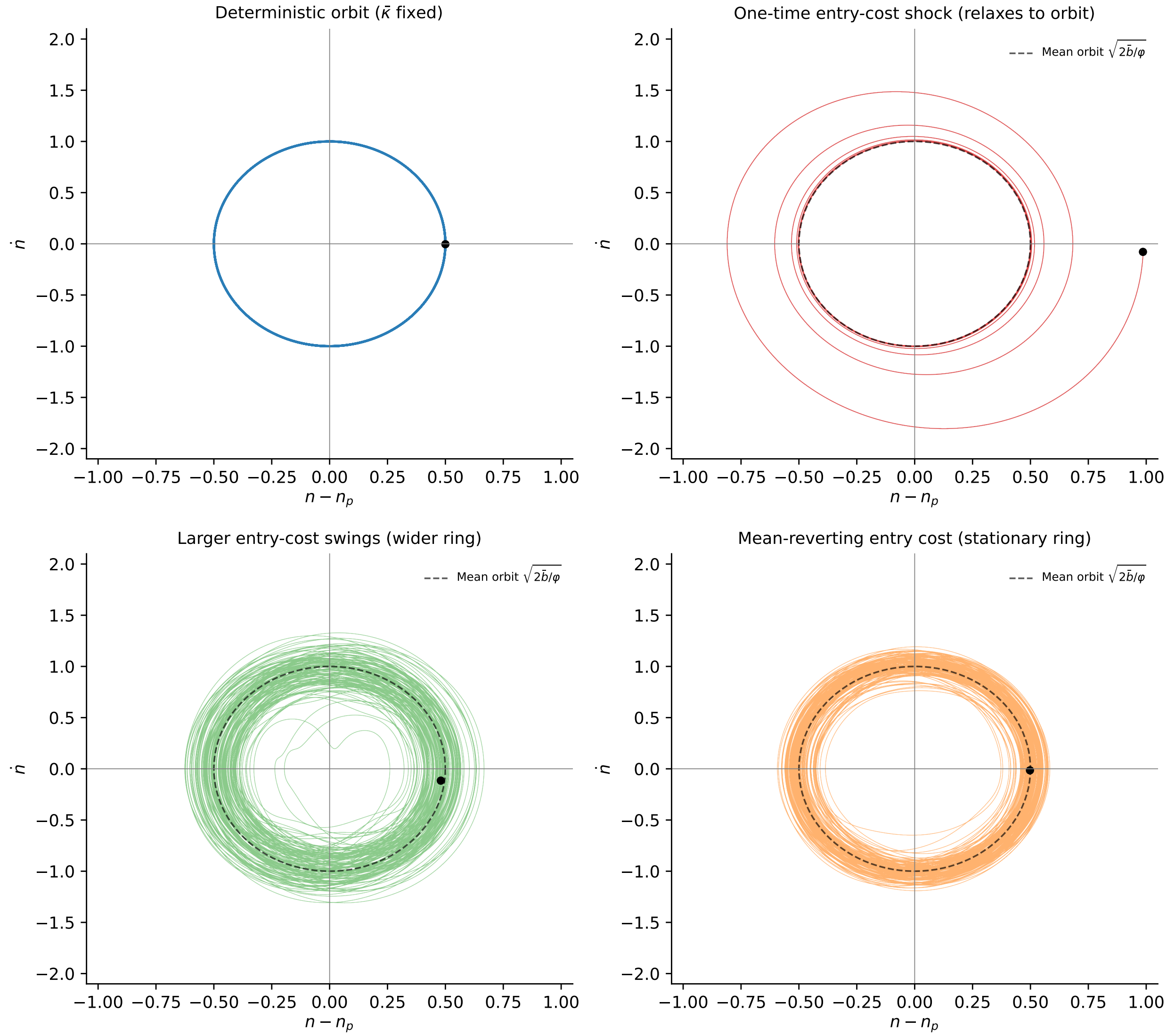
$$\mathbb{E}[\mathcal{L}] = \bar{b} + O(\mathrm{Var}(b)) = \ln \frac{\pi_0}{(r+\delta)\bar{\kappa}} + O\big(\tfrac{\sigma_\kappa^2}{2\theta}\big), \tag{19}$$

set by the entry-cost level $\bar{\kappa}$ and, at leading order, independent of $\mu$ (the second-order correction is derived in Section 5.1). A higher entry cost shrinks the cycle and its welfare cost, but raises the direct cost of entry; Section 5.1 works through the full trade-off.

The stochastic dynamics call for an equilibrium concept beyond the deterministic Definition 1, in which the value of entry fluctuates with the state rather than staying constant along the path.

**Definition 2** (Stationary timing search equilibrium)**.**
*The deterministic equilibrium of Definition 1 extends to this stochastic environment as follows. A* stationary timing search equilibrium *is a triple $(\mu^*, e^*, V^*)$. The triple consists of a*

Figure 1: The stationary cycle under fluctuating entry costs




*Notes:* Each panel plots the industry's position and speed of adjustment, $(n-n_p, \dot{n})$. Top left: the deterministic cycle at the mean entry cost $\bar{\kappa}$ (a closed loop). Top right: return after a one-time entry-cost shock (the amplitude reverts toward its long-run level $\sqrt{2\bar{b}/\varphi}$). Bottom left: wider amplitude variation under larger entry-cost swings. Bottom right: the stationary cycle under mean-reverting entry costs, with profit shortfall $b_t \sim \mathcal{N}(\bar{b}, \sigma_\kappa^2/2\theta)$.

*stationary distribution $\mu^*$ on $(n, \dot{n})$*[7] *with finite second moments, an entry policy $e^*$, and a value function $V^*$, satisfying:*

*(i)* ***Aggregate consistency****: under $e^*$, the stationary dynamics of this section admit $\mu^*$ as their invariant law, and $\dot{n}_t = e_t^* - \delta n_t$ along paths.*

[7] Here $(n, \dot{n})$ is the observable state, with $\dot{n} = e - \delta n$ the net entry rate. The stochastic state is the smooth deterministic cycle modulated slowly by $b_t$, so $\dot{n}_t$ is the genuine net entry rate (Online Appendix B.2).

*(ii)* ***Stochastic ironing (martingale form)****: at every $(n, \dot{n}) \in \operatorname{supp} \mu^*$, the value function $V^*$ satisfies the continuous-time Bellman equation*

$$\mathbb{E}\big[dV^*(n_t, \dot{n}_t) \bigm| \mathcal{F}_t\big] \;=\; \big[(r+\delta)\, V^*(n_t, \dot{n}_t) \;-\; \pi(n_t)\, \xi(\dot{n}_t)\big]\, dt. \tag{20}$$

*Equivalently, the discounted-value-plus-flow process*

$$e^{-(r+\delta)t}\, V^*(n_t, \dot{n}_t) \;+\; \int_0^t e^{-(r+\delta)s}\, \pi(n_s)\, \xi(\dot{n}_s)\, ds$$

*is a martingale under $\mu^*$. In the $\sigma_\kappa \downarrow 0$ limit, (20) reduces pathwise to the deterministic ironing condition of Proposition 1.*

*(iii)* ***No-deviation under stationary stopping****: for every $\mathcal{F}_t$-measurable stopping time $\tau$ whose marginal law satisfies $\mathcal{L}(n_\tau, \dot{n}_\tau) = \mu^*$, the expected option value $\mathbb{E}\big[V^*(n_\tau, \dot{n}_\tau)\big] = \mathbb{E}_{\mu^*}[V^*]$ is the same constant. No agent profits in expectation by retiming entry from one such stationary stopping rule to another. State-selecting rules with $\mathcal{L}(n_\tau, \dot{n}_\tau) \neq \mu^*$ are excluded from the comparison. They do not correspond to equilibrium entry strategies, which respect the stationary law. In the deterministic limit ($\sigma_\kappa \downarrow 0$), $\mu^*$ collapses to the deterministic cycle of Theorem 1 at amplitude $\sqrt{2\bar{b}/\varphi}$, and the condition reduces pathwise to the ironing condition $V(\tau) \equiv \kappa$ of Proposition 1.*

Among the infinite family of stationary distributions admissible under (i)–(ii), clause (iii) selects those at which conditional-expected option value is the same across the states the economy visits in the long run. In the linear-Gaussian benchmark of this paper, the Ornstein–Uhlenbeck dynamics already admit a single invariant law, so clause (iii) holds automatically; it becomes the binding restriction in non-Gaussian extensions where the deterministic landscape has multiple basins (Online Appendix B.3). Theorem 2 below confirms that exactly one such distribution exists. Under these linear-Gaussian dynamics, $V^*$ is a non-constant quadratic function of $(n - n_p, \dot{n})$, and the conserved object across stationary entry dates is the equilibrium value level $\mathbb{E}_{\mu^*}[V^*]$, not $V^*$ itself.

Proposition 4 pins the stationary variance of $(n_t - n_p)$ under the ironing condition. The next theorem strengthens that result by establishing two further properties: under a fluctuating entry cost, the stochastic dynamics of Section 3.6 admit a *unique* stationary timing search equilibrium in the sense of Definition 2, with amplitude distribution globally attracting from every initial law and phase uniform, and as shocks vanish the deterministic equilibrium of Theorem 1 re-emerges sharply as the limit.

**Theorem 2** (Existence and uniqueness of the stationary equilibrium)**.**

*Under Assumptions 1–4, in the small-fluctuation regime there is a unique stationary timing search equilibrium (Definition 2). The profit shortfall is stationary Gaussian,*

$$b_t \sim \mathcal{N}\Big(\bar{b},\ \tfrac{\sigma_\kappa^2}{2\theta}\Big), \qquad \bar{b} = \ln \frac{\pi_0}{(r+\delta)\bar{\kappa}},$$

*and its amplitude distribution is globally attracting at rate $\theta$ from any initial condition; the phase is uniform, the unique rotation-invariant law, approached in the time-average (ergodic) sense; the long-run averages used in the empirical work depend only on the amplitude distribution and the uniform phase. The cycle frequency is $\omega_0 = \sqrt{\varphi/\mu}$ regardless of the entry-cost process. As $\sigma_\kappa \downarrow 0$ the equilibrium collapses to the deterministic cycle of Theorem 1 at amplitude $\sqrt{2\bar{b}/\varphi}$.*

*Proof.* See Online Appendix B.3. The proof has three steps: the leading-order profit-shortfall law from the Ornstein–Uhlenbeck dynamics; uniqueness and convergence, the amplitude law attracting at rate $\theta$ and the phase uniform, the unique rotation-invariant law; and the vanishing-noise limit, which returns the deterministic cycle. ■

Theorem 2 supplies two things that Propositions 3–4 did not. First, *uniqueness* of the stationary distribution as an *equilibrium* object, not merely as the long-run limit of the dynamics. In the linear-Gaussian benchmark this follows from the Ornstein–Uhlenbeck law of $b_t$ having a single invariant distribution, and clause (iii) of Definition 2 is then satisfied automatically; it becomes substantive in non-Gaussian extensions (Online Appendix B.3). Second, a sharp deterministic-limit correspondence: as the entry-cost volatility vanishes ($\sigma_\kappa \downarrow 0$), the stationary cycle collapses to the deterministic cycle of Theorem 1 at amplitude $\sqrt{2\bar{b}/\varphi}$, pinned by the entry-cost level $\bar{\kappa}$. Section 3.7 uses the resulting frequency–amplitude–welfare structure to derive Theorem 3 in two lines.

**Shape and size of the stationary cycle** To leading order in the fluctuation, the industry is always on a cycle: it turns at $\omega_0$ around the loop set by the profit shortfall $b_t$, while $b_t$ varies slowly around $\bar{b}$. The stationary law is therefore a *fuzzy ring* in the $(n - n_p, \dot{n})$ plane, centred on the loop of radius $\sqrt{2\bar{b}/\varphi}$, rather than a cloud at the steady state. Its second moments still split cleanly: $\mathbb{E}[(n_t - n_p)^2] = \bar{b}/\varphi = \sigma_x^2$ measures how far the industry strays from optimal scale, and $\mathbb{E}[\dot{n}_t^2] = \bar{b}/\mu$ how fast it swings in and out, with zero covariance since deviation and rate are a quarter-cycle apart. The ring's *shape*, the ratio $\mathbb{E}[\dot{n}_t^2]/\mathbb{E}[(n_t - n_p)^2] = \varphi/\mu = \omega_0^2$, is fixed by the market-structure pair $(\varphi, \mu)$ alone, while its *size*, the mean profit shortfall $\bar{b} = \ln \frac{\pi_0}{(r+\delta)\bar{\kappa}}$, is fixed by the entry-cost level $\bar{\kappa}$. Rhythm lives in the shape, magnitude in the size, and the next subsection makes the disjointness exact.

### 3.7 Frequency-welfare separation

The closed-form structural frequency $\omega_0 = \sqrt{\varphi/\mu}$ (Theorem 1), the mean amplitude $\sigma_x = \sqrt{\bar{b}/\varphi}$ (Proposition 4, in which $\mu$ does not appear), and the welfare cost $\mathbb{E}[\mathcal{L}] = \bar{b} = \ln \frac{\pi_0}{(r+\delta)\bar{\kappa}}$ to leading order (Section 5.1) together establish the paper's central structural claim: *structural frequency and welfare cost have disjoint determinants*, with cycle amplitude standing as an intermediate observable depending on the overlapping pair $(\bar{\kappa}, \varphi)$. I state this as a theorem.

**Theorem 3** (Frequency-welfare separation)**.**
*In the timing search equilibrium under Assumptions 1–4 and Proposition 4, to leading order in the cycle amplitude:*

$$
\begin{aligned}
&\textit{frequency:} && \omega_0 = \sqrt{\varphi/\mu} && \textit{depends only on } (\varphi, \mu);\\
&\textit{amplitude:} && \sigma_x = \sqrt{\bar{b}/\varphi} && \textit{depends on } (\bar{\kappa}, \varphi);\\
&\textit{welfare cost:} && \mathbb{E}[\mathcal{L}] = \ln \tfrac{\pi_0}{(r+\delta)\bar{\kappa}} + O\big(\tfrac{\sigma_\kappa^2}{2\theta}\big) && \textit{depends on } \bar{\kappa} \textit{ at leading order.}
\end{aligned}
$$

*The determinant sets $(\varphi, \mu)$ and $\bar{\kappa}$ are disjoint, with cycle amplitude the intermediate observable connecting them.*

*Proof.* $\omega_0 = \sqrt{\varphi/\mu}$ is the structural frequency of the deterministic value landscape (Theorem 1); it depends only on $(\varphi, \mu)$. $\mathbb{E}[\mathcal{L}^*] = \bar{b}$ is the expected per-firm cycle-induced welfare cost under the unique stationary equilibrium $\mu^*$ (Theorem 2 combined with the welfare formula (19)); it depends only on $\bar{\kappa}$. The two are therefore pinned by disjoint coordinates of the primitive vector $(\varphi, \mu, \bar{\kappa})$ by construction. The cycle-amplitude formula $\sigma_x = \sqrt{\bar{b}/\varphi}$ is Proposition 4(ii), and the leading-order coincidence of the spectral peak with $\omega_0$ is Proposition 4(iii). Disjointness is preserved in the vanishing-volatility limit: as $\sigma_\kappa \downarrow 0$ the cycle limits to the deterministic cycle at amplitude $\sqrt{2\bar{b}/\varphi}$, with welfare cost $\bar{b}$ and frequency unchanged. ∎

The theorem has three methodological consequences. First, cross-industry variation in $\omega_0$ identifies the market-structure ratio $\varphi/\mu$ free of contamination from the entry-cost process; the cross-sectional regression of Section 4.3 therefore tests a structural prediction about $\varphi/\mu$ even when proxies for the individual primitives are imperfect. Second, two industries with identical $(\varphi, \mu)$ but different shock environments cycle at the same frequency, making the cross-industry frequency prediction structural rather than a reduced-form artifact of common shocks. Third, industrial policy targeting $\mu$ (entry regulation) or $\varphi$ (cluster development) affects timing alone; only policies that move the entry-cost level $\bar{\kappa}$ affect welfare (Section 5.3).

**Scope of the separation** Theorem 3's welfare claim is exact under the per-firm effective-profit welfare metric of (23); the full scope conditions and the single-digit median leakage (4–8%) under CRRA preferences, larger under metrics that charge the supplier-side resource flow, are stated in Section 5.1 and quantified in Online Appendix E.1.

**Comparison to alternative endogenous-cycle models** Timing search and Beaudry et al. (2020) share the broader insight that endogenous structure can turn short-lived disturbances into persistent cyclical adjustment, but Beaudry et al. (2020) generate a stochastic limit cycle through financial frictions and a Hopf bifurcation, while timing search generates characteristic frequencies from the entry-date indifference condition and a scale–congestion trade-off, which delivers the industry-decomposable closed form $\omega_0 = \sqrt{\varphi/\mu}$ tested in Section 4.3. Across standard benchmarks (AR(1) RBC, NK AR(2), multi-sector RBC with input-output linkages, and Khan and Thomas (2008)-style heterogeneous adjustment costs) the simulated cross-industry slope $\hat{\beta}_\gamma$ centers near zero with 95% CIs roughly $[-0.13, +0.20]$, none of 50 replications reaching the empirical $[0.50, 0.75]$ range; the comparison clarifies what additional structure the empirical question requires rather than ruling out enriched versions of these models (Online Appendices F.1–F.4).

# 4 Model vs. data

U.S. industries fluctuate at substantially different frequencies. Across six broad industry aggregates with bell-shaped interior peaks within the business-cycle band of 2–12 years (Burg AR(24) on monthly industrial production, 1972–2024), spectral peak periods range from 3.0 years (Nondurable Manufacturing) to 4.7 years (Business Equipment) (Section 4.2, Figure 3), a more than 50% spread across industries that share the same monetary policy, aggregate demand shocks, and financial conditions. Models that derive spectral properties from a common shock process or policy rule have difficulty reconciling this dispersion. Theorem 1's closed form $\omega = \sqrt{\varphi/\mu}$ predicts that the variation is *structural*: each industry's cycle frequency is pinned by market-structure primitives, not by the shock process. The remainder of this section organizes the empirical evidence around five questions the timing-search mechanism implies: whether industries cycle at different frequencies (Section 4.2); whether the structural slope $\beta_\gamma = 1/2$ holds (Section 4.3, the headline test); and three further diagnostics on whether the dynamics are genuinely oscillatory, whether the structural separation theorem holds across industries, and whether the model's predictions about cycle shape and recovery hold (Section 4.4).

## 4.1 Cross-industry predictions

Taking logs of $\omega_i = \sqrt{\varphi_i/\mu_i}$ yields the cross-industry regression framework

$$\ln \omega_i = \tfrac{1}{2} \ln \varphi_i - \tfrac{1}{2} \ln \mu_i + u_i, \tag{21}$$

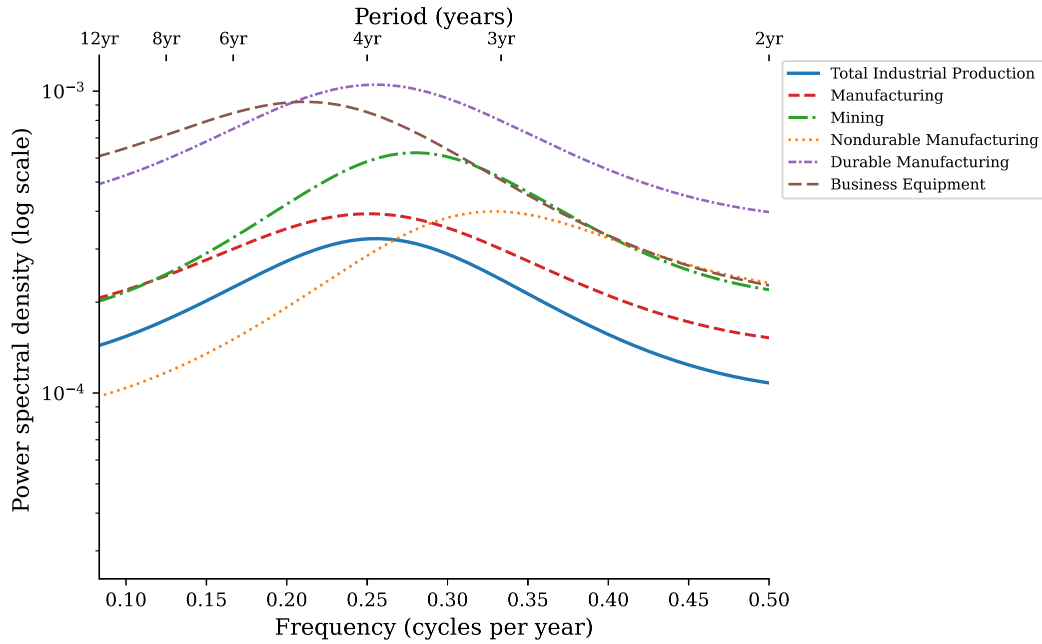
with the slopes $\pm 1/2$ as exact model predictions. The headline test of Section 4.3 restricts attention to 15 manufacturing 3-digit industries, within which the matching technology (permits, specialist construction labor, equipment supply chains) is approximately common across industries and the turnover rate $\delta_i \in [0.065, 0.090]$ varies only by a factor of 1.4. Under approximate $\mu$-uniformity, the prediction reduces to a $\varphi$-driven slope $\beta_\gamma = 1/2$. Because $\varphi_i$ is unobserved, the empirical implementation substitutes the geographic-concentration index $\gamma_i$ under a maintained measurement equation $\ln \varphi_i = a + \lambda \ln \gamma_i + v_i$, so the regression slope identifies $\lambda/2$, equal to 1/2 under unit log-elasticity ($\lambda \approx 1$); the evidence supporting this restriction is collected in Section 4.3. Two industries facing identical shock processes but differing in $(\varphi, \mu)$ are predicted to cycle at distinct frequencies; this prediction distinguishes the timing-search framework from RBC and New Keynesian frameworks, in which the spectral peak inherits its location from the shock process or the policy rule.

## 4.2 Spectral evidence from U.S. industry data

**Do industries cycle at different frequencies?** I evaluate the cross-industry prediction using monthly industrial production indices from FRED for eight U.S. sectors over 1972–2024. The series span a range of market structures: from Motor Vehicles & Parts (dense supplier networks, strong scale externalities) to Mining (dispersed extraction, weak inter-firm linkages) to Electric Power Generation (regulated, capital-intensive).

Computing spectral peaks via Burg AR(24) on monthly log-differenced production for each industry over 1972–2024 reveals substantial cross-industry variation within the business-cycle band of 2–12 years. Six of the eight FRED aggregates have interior bell-shaped peaks within the band (Figure 2): Business Equipment has the longest peak period at 4.7 years; Manufacturing aggregates and Mining cluster near 3.5–4.0 years (Total Industrial Production 3.9, Manufacturing 4.0, Durable Manufacturing 3.9, Mining 3.6); Nondurable Manufacturing peaks at 3.0 years. Two industries (Motor Vehicles & Parts and Electric Power Generation) concentrate spectral mass at the high-frequency boundary of the band, indicating sub-business-cycle dynamics, and are excluded from the cross-industry summary below. The dispersion across the six interior-peak industries is consistent with $\omega$ depending on market structure rather than on common shocks.

Figure 2: Spectral density of U.S. industry production growth (six interior-peak industries)

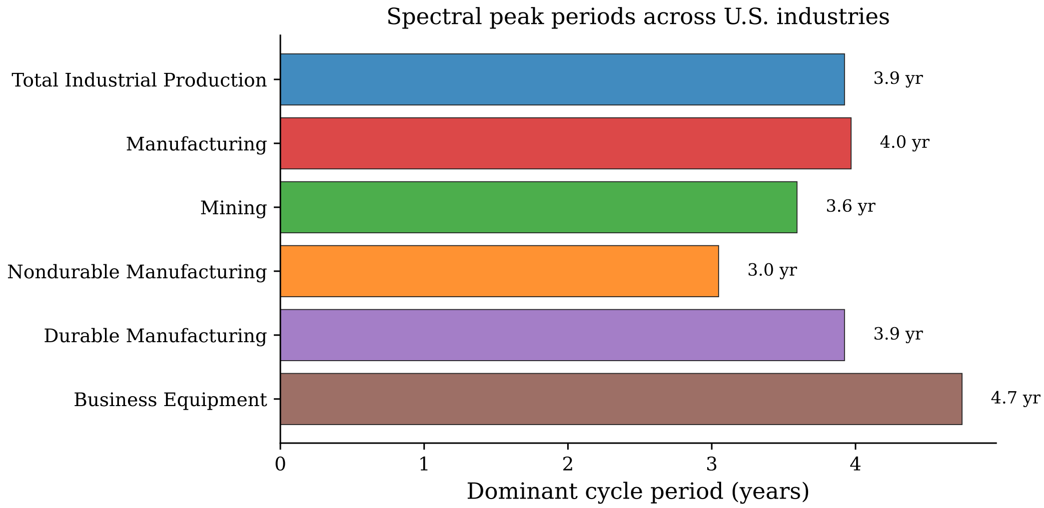



*Notes:* Burg AR(24) power spectral density of monthly log-differenced industrial production indices from FRED, 1972:M1–2024:M12, restricted to the business-cycle band of 2–12 years. The top axis labels convert frequency (cycles per year) to period (years). Each industry exhibits a bell-shaped interior peak within the band; peak periods range from 3.0 years (Nondurable Manufacturing) to 4.7 years (Business Equipment). Motor Vehicles & Parts and Electric Power Generation are excluded; their spectral mass concentrates at the high-frequency boundary, indicating sub-business-cycle dynamics.

Figure 3 summarizes the dominant cycle period for each of the six interior-peak industries. The cross-industry dispersion, ranging from 3.0 years to 4.7 years across industries that share a common macroeconomic environment, is difficult to reconcile with models that derive spectral properties from a common shock process. In the timing search framework, the variation is structural: it reflects cross-industry differences in market-structure primitives. Across this six-industry sample, which spans aggregate Manufacturing and Mining, both the profit-landscape curvature ($\varphi_i$) and the timing friction ($\mu_i$) plausibly vary across sectors with different matching technologies. The headline test of Section 4.3 narrows to 15 manufacturing 3-digit industries, within which $\mu$ is approximately uniform, so the cross-industry signal collapses to the $\varphi$ channel.

**Model shape versus data shape** The timing-search SDE generates an internal spectral peak at $\omega_0 = \sqrt{\varphi/\mu}$, while an AR(1) on growth rates produces a smooth low-pass spectrum with no internal peak. Figure 4 compares the model's simulated growth-rate spectrum at a manufacturing-like calibration to the FRED industrial-production growth spectrum for

Figure 3: Cross-industry variation in spectral peak periods




*Notes:* Dominant cycle period (years) for the six U.S. industries with interior bell-shaped spectral peaks within the business-cycle band of 2–12 years (Burg AR(24) on log-differenced monthly industrial production). Cross-industry peak periods range from 3.0 years (Nondurable Manufacturing) to 4.7 years (Business Equipment); manufacturing aggregates cluster near 3.5–4.0 years. Motor Vehicles & Parts and Electric Power Generation are excluded because their spectral mass concentrates at the high-frequency boundary, indicating sub-business-cycle dynamics.

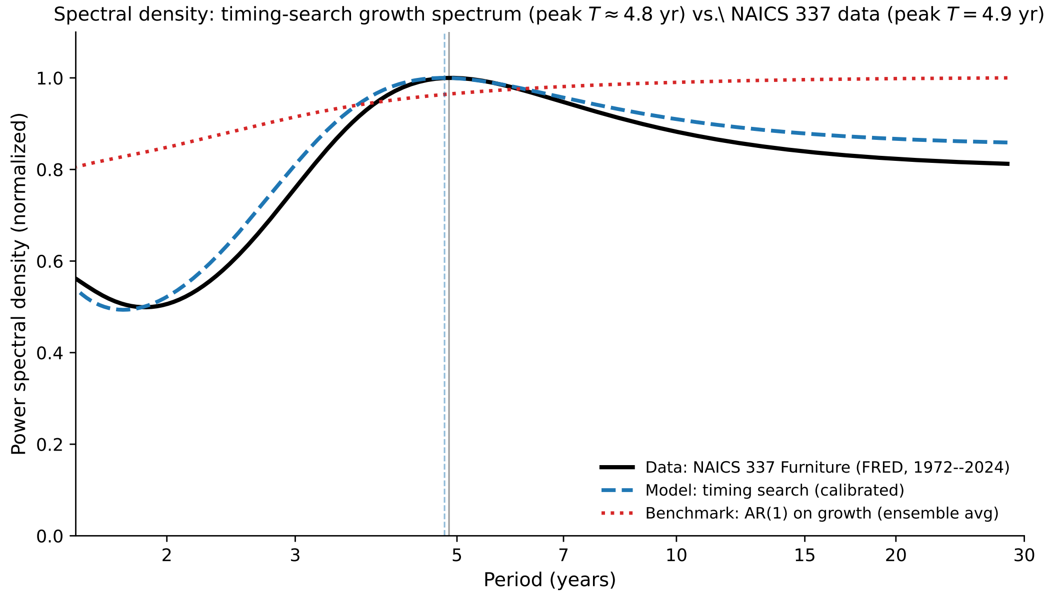
NAICS 337 (Furniture): both display a broad internal peak near $T = 4.9$ years of comparable width, while the AR(1) benchmark shows none. The model can produce a shape comparable to the data at a single industry, providing qualitative motivation for the cross-industry slope test below.

**Consistency check: nonlinear dependence in industry dynamics** As a consistency check rather than a discriminating test, the BDS test (Brock et al., 1996) rejects the i.i.d. null at the 1% level for all eight industries ($z$-statistics 16.2 to 21.8), confirming departure from linear AR(1) dynamics. The pattern is consistent with the timing-search mechanism but also with GARCH, regime switching, and structural breaks, so it does not discriminate among the nonlinear alternatives (full results, with AR(1) and AR(p)+GARCH pre-filtering, in Online Appendix D.2).

## 4.3 Cross-industry regression with external proxies

**Does the structural slope $\beta_\gamma = 1/2$ hold under the maintained restrictions?** The cross-industry regression tests the closed-form prediction (21) under three explicit maintained restrictions, each disclosed up front rather than as a footnote: (i) approximate $\mu$-uniformity across 3-digit manufacturing industries, with the observable $\delta$-driven component partialled

Figure 4: Spectral density: timing-search model vs. industry data




*Notes:* Burg AR(24) power spectral densities, normalized to the maximum within the business-cycle band (1.5–30 years). Solid black: FRED monthly industrial production growth for NAICS 337 (Furniture), 1972–2024, with empirical peak at $T = 4.9$ years. Dashed blue: simulated timing-search growth-rate spectrum at a manufacturing-like calibration ($\varphi = 1.28$, $\mu = 1$, $\theta = 0.40$, $\sigma_\kappa = 0.04$; level structural frequency $T_0 = 2\pi/\sqrt{\varphi/\mu} = 5.5$ years), ensemble-averaged over 30 replications, with realistic-conditions noise components (idiosyncratic measurement noise $\sigma_{\text{meas}} = 0.045$ and a slow AR(1) productivity-drift component, $\rho = 0.93$, $\sigma = 0.0011$) that mimic the finite-sample data generating process. The calibration is chosen to match the data on both the peak location and the dispersion: the model and data growth-rate spectra share the same broad half-power band (roughly 1.5–28 years). Both curves are log-difference (growth-rate) spectra; differencing high-passes the level spectrum, so the growth-rate peak sits below the level frequency $T_0$ by a factor common to model and data (it cancels in the cross-industry slope of Section 4.3), landing near $T = 4.9$ years for both. Dotted red: ensemble average of 30 AR(1)-on-growth-rate replications ($\rho = 0.5$), shown for comparison, a standard "no internal peak" benchmark with smooth low-pass spectrum. Timing search can generate a comparable peak and surrounding spectral shape; AR(1) cannot.

out as a robustness specification below and Online Appendix D.5; (ii) a measurement equation $\ln \varphi_i = a + \lambda \ln \gamma_i + v_i$ that maps the unobserved profit-landscape curvature to the Ellison and Glaeser (1997) concentration index, so the empirical slope identifies $\lambda/2$, equal to $1/2$ at unit log-elasticity ($\lambda = 1$); (iii) the timing-search mechanism's harmonic-oscillator structure dominates at the spectral resolution targeted by the multitaper and Burg AR(12) estimators, which identify the dominant low-frequency mode. Each restriction is either tested directly (i, via the $\mu$-channel three-part defense), disciplined by the revealed-curvature $\lambda$ diagnostic below while still maintained as an auxiliary link (ii), or transparently dependent on a methodological choice (iii). Under all three, the test is the cross-industry structural-prediction test; outside any of the three, the same regression is a consistency diagnostic for an enriched model with a $\gamma$-correlated $\mu$, a non-unit $\lambda$, or a broader spectral target.

A broad-sector baseline (seven BEA industry groups, inverse I-O self-purchase for $\varphi$ and capital per employee for $\mu$) gives a univariate slope of 0.85 ($R^2 = 0.65$, $p = 0.028$; Online Appendix D.1). The headline test moves to $N = 15$ three-digit manufacturing industries with the Ellison and Glaeser (1997) geographic concentration index $\gamma_i$ as the $\varphi$ proxy and multitaper (Thomson DPSS, $K = 5$) spectral peaks as $\omega_i$, with Burg AR(12) and 2SLS as robustness specifications.[8]

*Estimator hierarchy.* The multitaper spectral estimator (Thomson DPSS, $K = 5$) is the primary one throughout and Burg AR(12) the same-band robustness check; both identify the dominant low-frequency peak the harmonic-oscillator structure maps to. The higher-order Burg AR(24) and the Welch periodogram over-resolve the business-cycle band at NAICS-3 resolution and are reported as sensitivities (Online Appendix D.6).

Table II reports the slope under several specifications. The cross-industry $\omega_i$ measure is the multitaper spectral peak (Thomson DPSS, $K = 5$ tapers) for the headline specifications; column 2 reports a Burg AR(12) robustness check on the spectral side. Across these specifications the empirical slope spans $[0.50, 0.75]$, all within roughly one HC1 standard error of the structural prediction $\beta_\gamma = 1/2$ from (21). The multitaper estimator gives $\hat{\beta}_\gamma = 0.50$ ($p = 0.004$, $R^2 = 0.56$); Burg AR(12) on the same series gives $\hat{\beta}_\gamma = 0.53$ ($p = 0.029$); the strong-first-stage broad-1947 IV gives $\hat{\beta}_\gamma^{\mathrm{IV}} = 0.53$; and the primitive Kim+CBP instrument set gives $\hat{\beta}_\gamma^{\mathrm{IV}} = 0.75$. Wald and Anderson-Rubin tests fail to reject $\beta_\gamma = 1/2$ at conventional levels. The sign matches theory: more geographically concentrated industries, those with stronger agglomeration externalities and higher $\varphi$, cycle faster.

*Higher-order Burg AR(24).* The higher-order Burg AR(24) gives a lower $\hat{\beta}_\gamma = 0.24$. The divergence is not a finite-sample artifact (AR(1)-, AR(2)-, and BIC-prewhitened variants all give 0.22–0.24) but over-resolution of the business-cycle band, which shifts long-cycle industries to higher frequency and compresses cross-industry $\omega$ variance (Online Appendix D.6); the structural test is read off the multitaper and AR(12) specifications.

*Historical-geography 2SLS.* The main IV instruments modern $\gamma_i$ with a single 1947 broad-regional specialization score $Z^B_{47,i}$, combining Kim's state-level Hoover localization coefficient (Kim, 1995) with state and county concentration from the digitized County Business Patterns of Eckert et al. (2022) (construction in Online Appendix D.5). It yields $\hat{\beta}_\gamma^{\mathrm{IV}} = 0.53$ (HC1 SE 0.22, $N = 15$) with first-stage $F_{\mathrm{HC1}} = 17.81$. Wald (1943) inference fails to reject $\beta_\gamma = 0.5$ ($p = 0.89$); Anderson and Rubin (1949) weak-IV-robust testing gives $p = 0.036$ against zero, and the inverted 95% AR confidence set is $[+0.06, +1.09]$, bounded and connected,

[8] The proxy swap is empirical: at the 3-digit level the broad-sector I-O and capital-intensity proxies do not preserve the relationship, while the EG index, constructed at 3-digit manufacturing resolution by design, does (Online Appendix D.1).

a first-stage signature rather than the unbounded set a genuinely weak instrument would produce, though the exclusion of zero is marginal. The honest reading is that the IV is corroborative on sign and order of magnitude rather than a precise test of the structural value: the argument runs through the joint reading of the closed-form prediction (21), the OLS points near 0.5, and the IV's positive bounded set, not a pointwise IV t-ratio that the $N = 15$ design cannot deliver.

Primitive IV constructions (Kim 1947 alone, Kim+CBP county Herfindahl, Kim+1947 firm-level CR4) span $[0.53, 0.75]$ across five specifications with Hansen $J$ never rejected, and an instrument-design scan confirms the broad-1947 score clears $F_{\text{HC1}} > 10$ in the full, strict-band, and every leave-one-out sample (Online Appendix D.5).

The exclusion restriction rests on pre-1947 geography pre-dating the modern cycle environment; the most substantive threat, a path from 1947 geography through persistent regulation into $\mu$, is tested directly and not supported (Online Appendix D.5). The IV result is best read alongside the multitaper and AR(12) OLS estimates as evidence pointing to $\beta_\gamma$ near the structural 1/2 of (21).

*Revealed-curvature diagnostic for $\lambda$.* A revealed-curvature diagnostic, regressing the model-implied $\ln \hat{\varphi}_i^\delta \propto \ln(\omega_i/\delta_i)^2$ on $\ln \gamma_i$, gives $\hat{\lambda} \approx 1.0$ for the headline estimators and does not reject $\lambda = 1$, so the unit-elasticity link is consistent with the data conditional on the structural equation, though not an independent measurement (Online Appendix D.5).

*Structural reading of the slope.* The slope on $\ln \gamma_i$ equals 1/2 when $\ln \mu_i$ is approximately uncorrelated with $\ln \gamma_i$ and the proxy elasticity $\lambda \approx 1$. Within manufacturing this is plausible: the matching technology (permits, specialist labor, equipment supply chains) is generic to the entry pipeline, and the structural friction $\tilde{\mu}_i = (1 - \alpha)/(\alpha \delta_i^2)$ varies narrowly over the $1.4\times$ range of $\delta_i$, while $\gamma$ varies $30\times$ (0.004 to 0.12), so cross-industry frequency variation operates through the profit-landscape curvature $\varphi$. Controlling explicitly for the observable $\ln \delta_i$, rather than assuming $\mu$ uniform, leaves $\hat{\beta}_\gamma$ essentially unchanged (OLS $0.497 \to 0.495$; broad-1947 IV $0.531 \to 0.532$), with the $\delta$-channel itself unidentified at that narrow range (Online Appendix D.5).

Three external proxies support $\mu$-uniformity-on-$\gamma$ (Online Appendix D.5): two capital-side proxies (the BEA replacement horizon and the Cooper and Haltiwanger (2006) adjustment-cost coefficient) are indistinguishable from zero on $\ln \gamma_i$, while a QuantGov RegData regulatory-restriction count is positive ($\hat{\beta} = +0.91$, $p = 0.07$). Under the model's sign convention regulation enters $\mu$ positively, so the omitted $-\frac{1}{2} \ln \mu_i$ biases the OLS slope *toward* zero, making the recovered $[0.50, 0.75]$ a lower bound on the $\varphi$-channel slope; Frisch-Waugh orthogonalization on the regulatory proxy leaves the slope within one HC1 SE of 1/2. The historical instrument is empirically orthogonal to the regulatory contaminant ($\ln R_i$ on $Z^B_{47,i}$:

$\hat{\beta} = +0.20$, $p = 0.83$), so the path 1947 geography → regulation → $\mu \to \omega$ is not supported, and Hansen-$J$ overidentification is never rejected.

Table II: Cross-industry frequency regression

| | OLS MT $N = 15$ | OLS AR(12) $N = 15$ | 2SLS Broad $N = 15$ | 2SLS Kim+CBP $N = 15$ | Theory |
|---|---|---|---|---|---|
| $\hat{\beta}_\gamma$ | 0.50 | 0.53 | 0.53 | 0.75 | 0.50 |
| HC1 SE | (0.14) | (0.22) | (0.22) | (0.24) | — |
| Wald $p$ vs. 0 | 0.004 | 0.029 | 0.034 | 0.008 | — |
| Wald $p$ vs. 0.5 | 0.98 | 0.89 | 0.89 | 0.32 | — |
| AR $p$ vs. 0 | — | — | 0.036 | 0.015 | — |
| AR $p$ vs. 0.5 | — | — | 0.87 | 0.36 | — |
| Hansen $J$ $p$ | — | — | — | 0.40 | — |
| $R^2$ / 1st-stage F | 0.56 | 0.30 | $F = 17.81$ | $F = 7.29$ | — |

*Notes:* Cross-industry regression of $\ln \omega_i$ on $\ln \gamma_i$ (Ellison and Glaeser 1997 geographic concentration index). Column 1: OLS with multitaper spectral peaks (Thomson DPSS, $K = 5$, Thomson 1982). Column 2: OLS with Burg AR(12) spectral peaks (different $\omega_i$ measure as a spectral-side robustness check). Column 3: 2SLS of multitaper $\omega_i$ on $\gamma_i$ using the broad 1947 regional-specialization score $Z^B_{47}$ as a single excluded instrument. Column 4 uses the primitive Kim 1947 Hoover and 1947 CBP county-level employment Herfindahl as two excluded instruments; the reported first-stage statistic is the HC1-Wald joint statistic for the excluded instruments. Spectral peaks estimated from log-differenced monthly industrial production (FRED, 1972–2024). HC1 heteroskedasticity-robust standard errors (White, 1980). AR rows report Anderson and Rubin (1949) weak-IV-robust $p$-values for the IV columns. The theory column is conditional on the auxiliary unit-elasticity link $\lambda = 1$ in $\ln \varphi_i = a + \lambda \ln \gamma_i + v_i$; the revealed-curvature diagnostic in the text estimates $\hat{\lambda} \approx 1.0$ for the headline multitaper and AR(12) specifications. The broad-1947 score also clears the conventional homoskedastic first-stage threshold ($F = 11.32$) and the strict-band/leave-one-out first-stage checks reported in Online Appendix D.5. The higher-order Burg AR(24) yields $\hat{\beta}_\gamma = 0.24$ (Wald $p = 0.002$ vs. 0.5), the only specification in the Online Appendix D.6 six-estimator panel that rejects 0.5 at the 5% level; the Welch periodogram falls at 1.9 HC1 SE below 0.5 ($p \approx 0.07$) and is reported with AR(24) as a smaller-slope, same-sign sibling.

Two further exercises corroborate the cross-section without overturning it: an $N = 10$ BDS-subsector cross-section using a BDS-derived concentration measure gives the same-sign $\hat{\beta} = 0.107$ ($p = 0.037$), and a four-digit exercise collapses mechanically because Ellison et al. (2010) publish $\gamma$ only at three-digit resolution (a US+UK pooled exercise is same-signed but proxy-limited; Online Appendix D.4, D.7).

**Identification** The ratio $\varphi/\mu$ is identified directly from the spectral peak ($\varphi/\mu = (2\pi/T^*)^2$, exactly). Individual $\varphi$ and $\mu$ require external information; $\bar{\kappa}$ is identified from the cycle amplitude via $\bar{b} = \varphi\sigma_x^2$ (so $\bar{\kappa} = \pi_0 e^{-\varphi\sigma_x^2}/(r + \delta)$), and the persistence $\theta$ from the amplitude-decay rate.

## 4.4 Further diagnostics

The cross-sectional regression in Section 4.3 is the paper's structural test, identifying the $\varphi$ channel through the geographic-concentration proxy. I complement it with three further empirical questions, each addressing a distinct prediction of the timing-search mechanism beyond the cross-industry slope. I present these as suggestive rather than definitive evidence; their joint pattern is harder to reproduce in any single alternative model than in any one diagnostic taken alone.

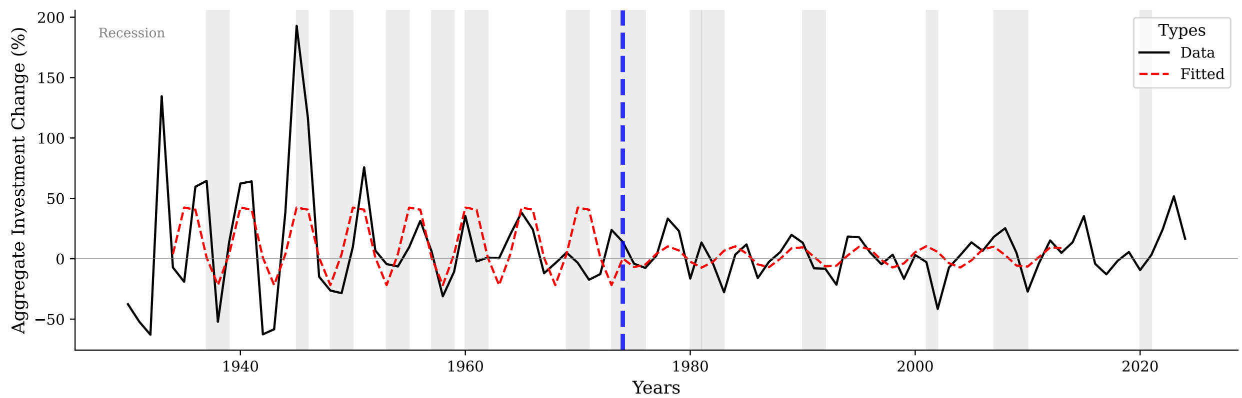
**Are the dynamics genuinely oscillatory rather than persistent AR(1)?** *Oscillatory entry dynamics across industries (BDS, 1978–2022).* For each of 19 NAICS sectors, the linearly detrended net entry rate exhibits two features outside the AR(1) class: four sectors show significantly negative lag-4 autocorrelation outside the 95% Bartlett (1946) band (incompatible with any scalar AR(1)), and Fisher's $g$-test rejects the null of no deterministic periodicity at the 5% level in 10 of 19 sectors, with peaks near $T \approx$ 8–9 years. The sign-changing autocorrelation pattern is also harder to reconcile with vintage-capital replacement, which produces positively-correlated entry waves rather than the lag-4 sign reversal documented above. Online Appendix D.2 provides the full sector-by-sector results.

*Post-shock periodicity in historical impulse responses.* I use BEA investment-in-structures data (1930–2024) to test whether one-time shocks trigger oscillations at industry-specific structural frequencies. Manufacturing investment after the Great Depression onset oscillates at $T = 5.0$ years through the early 1970s (Figure 5a; $g$-test $p = 0.036$ in the 1934–1973 echo window, $p = 0.578$ in 1974–2013), consistent with the model's prediction of decaying post-shock oscillation at fixed period. Mining/oil investment after the 1979 crisis oscillates at $T = 3.25$ years (Figure 5b; $p = 0.016$). The cross-industry frequency difference is consistent with different $(\varphi, \mu)$ across industries (Online Appendix D.3).

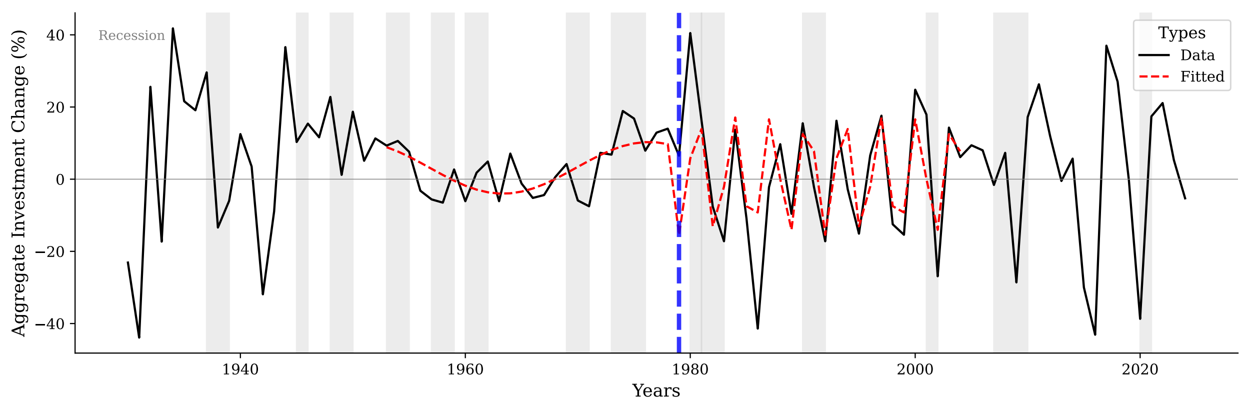
**Does the structural separation theorem hold across industries?** *Three-moment fingerprint of frequency-welfare separation.* Theorem 3 predicts that the per-industry trio $(\omega_{0,i}, \theta_i, \sigma_{x,i})$ is governed by disjoint primitive subsets: $\omega_{0,i}^2 = \varphi_i/\mu_i$, the entry-cost persistence $\theta_i$, and $\sigma_{x,i}^2 = \bar{b}_i/\varphi_i$. Under approximate $\mu$-uniformity, cross-industry variation in $\omega_0$ should be uncorrelated with $\theta$ and negatively correlated with $\sigma_x$, provided $\bar{b}$ does not have offsetting positive covariance with $\varphi$. I fit an AR(2) by Yule-Walker to the band-pass-filtered detrended log level for $N = 15$ NAICS-3 manufacturing industries, reading the entry-cost persistence $\theta_i$ off the fitted amplitude-decay rate (Online Appendix D.8). The separation theorem predicts $\rho(\omega_0, \theta) = 0$. A single-channel persistence model, in which one parameter governs both period and decay, instead predicts strong positive $(\omega_0, \theta)$ co-movement. The data give

$\rho(\omega_0, \theta) = -0.23$ ($p = 0.41$). The point estimate is consistent with separation, but the test does not have power to reject the single-channel alternative at $N = 15$. The remaining pairwise correlations are $\rho(\omega_0, \sigma_x) = -0.52$ ($p = 0.05$) and $\rho(\theta, \sigma_x) = -0.16$ ($p = 0.57$). The negative $\omega_0$-$\sigma_x$ correlation is consistent with $\mu$-uniformity, subject to the offsetting-covariance caveat.

Figure 5: Post-shock periodicity: manufacturing vs. mining/oil investment




(a) Manufacturing structures




(b) Mining exploration, shafts, and wells

*Notes:* Annual growth rate of private fixed investment from BEA NIPA Table 5.4.1: panel (a) manufacturing structures (line 16), panel (b) mining exploration, shafts, and wells (line 24). Thick solid line: data. Red dashed line: fitted cosine from Fisher's $g$-test. Gray bands: NBER recessions. Blue dashed lines mark structural breaks: in (a), the boundary between the post-shock window 1934–1973 and the non-echo window 1974–2013; in (b), the 1979 oil crisis. Manufacturing oscillates at $T = 5.0$ years from the Great Depression onset through the early 1970s ($p = 0.036$); mining/oil oscillates at $T = 3.25$ years post-1979 ($p = 0.016$), consistent with different $(\varphi, \mu)$ across industries.

**Do the timing-search predictions about cycle shape and recovery hold?** *The post-COVID natural experiment.* Timing search predicts displaced industries overshoot the pre-shock level rather than converge monotonically. The ironing condition forces the economy past optimal scale before cycling back. After the 2020 shock, 12 of 19 BDS NAICS sectors (63%) overshoot above their pre-COVID linear trend in 2021 or 2022 by at least one pre-COVID standard deviation. The corresponding AR(1)-monotone-convergence baseline is 0.26 (binomial test $p < 0.001$). To compute the baseline, I simulate a sector-specific AR(1) fitted on the linearly detrended pre-COVID net entry rate (1978–2019), starting from the realized 2020 deviation. I then record the share of 10,000 replications whose 2021–2022 path rises at least one pre-COVID standard deviation above the pre-shock trend purely from innovation noise (Online Appendix D.9 specifies the procedure). The qualitative prediction is supported; the quantitative prediction that overshoot magnitude correlates with concentration is not, with sector-specific COVID exposure dominating cross-sectional variation.

*Plucking pattern.* A depth-independent period plus amplitude-proportional recovery speed (Section 3.4) implies the recovery rate scales with recession depth while recovery time does not. For each of seven NBER recessions in FRED INDPRO, 1972–2024, I regress average monthly log growth in the first 12 months post-trough on the peak-to-trough log decline. The slope is +0.046 ($p = 0.029$, $R^2 = 0.65$, Pearson $r = +0.80$). Deeper recessions are followed by faster recoveries, the timing-search analogue of the plucking pattern documented in Friedman (1993); Bordo and Haubrich (2017) and Dupraz et al. (2019).

# 5 Welfare and industrial policy

## 5.1 Welfare cost of endogenous cycles

Under the linear-utility household specification of Section 2, household consumption equals aggregate output, so the household welfare loss from cycling is $n_p\, \pi_0\, \rho^{-1}\, \mathbb{E}[\mathcal{L}]$ to leading order, where $\mathcal{L}$ is the proportional per-firm output deviation. I use "per-firm welfare cost" as shorthand for $\mathcal{L}$ throughout.

The ironing condition provides a natural welfare measure. Along any equilibrium cycle, the profit shortfall $b = \frac{\varphi}{2}A^2$ is constant by the ironing condition (Equation (10)), and the time-averaged effective profit is:

$$\bar{y} = \frac{1}{T}\int_0^T \pi(n_t)\, \xi(\dot{n}_t)\, \mathrm{d}t = C = \pi_0\, e^{-b} = \pi_0\, \exp\Big(-\frac{\varphi}{2}A^2\Big)\,. \tag{22}$$

Since $A > 0$ in any stochastic equilibrium, $\bar{y} < \pi_0 = \pi(n_p)\xi(0)$: the time-averaged per-firm effective profit is strictly below the optimal-scale, zero-adjustment level. The proportional per-firm effective-profit welfare loss from cycling is:

$$\mathcal{L} = 1 - \frac{\bar{y}}{\pi_0} = 1 - e^{-\varphi A^2/2} \approx \frac{\varphi}{2} A^2 \quad \text{for small } A. \tag{23}$$

Taking expectations over the stationary profit shortfall $b \sim \mathcal{N}(\bar{b}, \mathrm{Var}(b))$ with $\mathrm{Var}(b) = \sigma_\kappa^2/2\theta$,

$$\mathbb{E}[\mathcal{L}] = 1 - \mathbb{E}[e^{-b}] = 1 - e^{-\bar{b}+\mathrm{Var}(b)/2} = \bar{b} + O(\mathrm{Var}(b)), \tag{24}$$

so the mean welfare cost equals the mean profit shortfall $\bar{b} = \ln \frac{\pi_0}{(r+\delta)\bar{\kappa}}$ *to leading order* in the cycle amplitude. The second-order term carries the entry-cost volatility and persistence through $\mathrm{Var}(b) = \sigma_\kappa^2/2\theta$; because $\mathcal{L}$ is concave in $b$ it is negative, so the separation of welfare from $(\sigma_\kappa, \theta)$ is a leading-order property, exact only for the log-loss metric $\mathbb{E}[\ln(\pi_0/\bar{y})] = \mathbb{E}[b] = \bar{b}$. The welfare cost depends on two structural parameters. The first is $\varphi$, the curvature of the profit landscape, which determines how costly deviations from optimal scale are. The second is $A$, the cycle amplitude: on a cycle with profit shortfall $b$ the peak amplitude is $A = \sqrt{2b/\varphi}$, set in stochastic equilibrium by the entry-cost level $\bar{\kappa}$ and the profit-landscape curvature $\varphi$, with root-mean-square counterpart $\sigma_x = \sqrt{\bar{b}/\varphi} = A_{\text{mean}}/\sqrt{2}$. The welfare loss $\mathcal{L}$ does not depend on $\mu$. The timing friction determines the *frequency* of cycles but not their welfare cost. Two industries with the same $\varphi$ and $A$ but different $\mu$ cycle at different frequencies yet suffer identical welfare losses.

This gives the Lucas (1987) question a structural answer. Lucas takes the variance of consumption as a reduced-form object. Here, to leading order $\mathbb{E}[\mathcal{L}] \approx \ln \frac{\pi_0}{(r+\delta)\bar{\kappa}}$ is a function of measurable primitives: the entry-cost level $\bar{\kappa}$ and the peak profit $\pi_0$. In the linear small-amplitude limit, the welfare cost is independent of cycle frequency (Theorem 3). Online Appendix E.1 derives the welfare metric and isolates the supplier-side resources the entry pipeline spends absorbing the cycle as a separate accounting object.

**Scope of the disjoint-determinants result** The disjointness $\mathbb{E}[\mathcal{L}] = \ln \frac{\pi_0}{(r+\delta)\bar{\kappa}}$ is exact for the log-loss metric $\mathbb{E}[b] = \bar{b}$ and holds to leading order in the cycle amplitude under the per-firm effective-profit metric of (23); the second-order term $\mathrm{Var}(b)/2 = \sigma_\kappa^2/4\theta$ reintroduces the entry-cost volatility and persistence. Under CRRA preferences, an aggregate-consumption-equivalent metric, or a welfare measure that internalizes the supplier-side resource cost, the primitives $(\varphi, \mu)$ also enter. Online Appendix E.1 quantifies these as 4–8% at the median under CRRA preferences, and larger under a metric that also charges the supplier-side resource flow (which reintroduces $\mu$). The qualitative ranking is preserved: market-structure primitives govern timing, while the entry-cost level governs the leading-order welfare. The CHIPS/IRA counterfactual in Section 5.3 is therefore a leading-order statement, with the second-order and metric corrections bounding the robustness range.

Calibrating $\bar{b}_i = \varphi_i \sigma_{x,i}^2$ for each manufacturing industry yields a mean welfare cost of 0.39% of $\pi_0$ across 10 cached industries, with a range of $[0.12\%, 0.70\%]$ (Online Appendix E.2). These magnitudes are comparable to Lucas (1987)- and Reis (2009)-style consumption-equivalent business-cycle welfare costs. The welfare numbers should be read as central estimates with the same precision caveats that apply to the slope test. The model-implied second-order fit identifies the structural frequency $\omega_0^2 = \varphi/\mu$ and the entry-cost persistence $\theta$, and the welfare cost $\mathbb{E}[\mathcal{L}] = \bar{b} = \varphi\sigma_x^2$ reads directly off the fitted amplitude and curvature, with no auxiliary $\mu$ calibration. With $\sim$44 annual BDS observations per industry, the per-industry amplitude $\sigma_x$ carries small-sample uncertainty. The cross-industry mean of 0.39% carries only small-sample uncertainty in the fitted amplitudes (Online Appendix E.2). They should be read in the same spirit as the $N = 15$ slope test's wide AR confidence set, rather than as point identification.

## 5.2 The cycle as an entry externality

The welfare cost $\bar{b}$ admits a sharper reading: it is the entry surplus that free entry dissipates through inefficient overshooting, and a corrective instrument removes it. Two externalities drive the overshoot, both already present in the environment of Section 2. A searcher who enters raises the active mass $n$, shifting every incumbent's profit by $\pi'(n)$, which is negative past $n_p$ (the agglomeration externality); and she adds to the net entry rate $\dot{n}$, congesting the shared matching process and lowering $\xi(\dot{n})$ for every contemporaneous searcher, since $\xi'(\dot{n}) < 0$ away from $\dot{n} = 0$ (the congestion externality). The atomistic firm internalizes neither: it equalizes its own flow payoff $\pi(n)\xi(\dot{n})$ across dates, taking the aggregate path as given.

A planner who internalizes both does not cycle. Maximizing household consumption $\int_0^\infty e^{-\rho t}\left[n_t\pi(n_t)\xi(\dot{n}_t) - \kappa e_t\right] \mathrm{d}t$ subject to $\dot{n}_t = e_t - \delta n_t$, the planner's optimum is a rest point ($\dot{n} = 0$) at the scale $n^*$ solving

$$\pi(n^*) + n^*\,\pi'(n^*) = (\rho + \delta)\,\kappa, \tag{25}$$

which equates the marginal aggregate profit of an entrant to the entry-cost annuity (Online Appendix E.4). The planner does not oscillate because $\xi(\dot{n}) \leq 1$ with equality only at $\dot{n} = 0$ and $\pi$ is concave around $n_p$, so any cycle strictly lowers time-averaged output; as the entry surplus vanishes, $n^* \to n_p$. The decentralized ironing condition $\pi(n)\xi(\dot{n}) = (r + \delta)\kappa$ omits the agglomeration term $n\pi'(n)$ and the congestion carried by $\xi'$; with the household owning firms so that $r = \rho$, the gap between the two conditions is exactly the uninternalized externality.

**Proposition 5** (Corrective instrument)**.**
*An entry tax $\tau$, rebated lump-sum to the household, implements the no-cycle allocation. The flat fee $\tau = \pi_0/(r+\delta) - \kappa$ raises the effective entry cost until the ironing locus $\pi(n)\xi(\dot{n}) = (r+\delta)(\kappa+\tau) = \pi_0$ collapses to the single state $(n_p, 0)$, so the cycle disappears and $\bar{b} \downarrow 0$. This coincides with the planner's rest point $n^*$ of (25) to leading order in the entry surplus (they agree as $\bar{b} \to 0$); because the tax is a transfer, household welfare rises by the eliminated loss $n_p \pi_0 \rho^{-1} \bar{b}$.*

This sharpens the welfare reading of Section 5.1. The handle on the welfare cost is the *effective* entry wedge, and the efficient way to move it is a corrective tax: a transfer that captures the surplus the cycle would otherwise burn and returns it, rather than a higher real cost. Raising the *real* entry cost $\bar{\kappa}$ reaches the same $\bar{b}$ but spends resources instead of rebating them, so it is dominated by the tax and only weakly welfare-improving once the supplier-side resource cost is charged (Online Appendix E.1). The policy reading inverts accordingly: the model calls not for higher entry barriers but for a Pigouvian tax on the timing of entry, and capacity-expansion subsidies, which cut the entry wedge, are the anti-corrective move that enlarges the inefficient cycle (Section 5.3).

## 5.3 Industrial policy and the anatomy of recovery

The frequency-welfare separation (Theorem 3) delivers an instrument-to-outcome map under the per-firm effective-profit welfare metric. The structural parameters act on identifiable margins: $\mu$ governs timing alone, $\varphi$ governs timing and amplitude, the entry-cost level $\bar{\kappa}$ governs amplitude and welfare, and the entry-cost volatility $\sigma_\kappa$ and persistence $\theta$ govern the cyclical fluctuations but not the mean welfare to leading order (they enter only at second order, through the convexity of the loss; Section 5.1). In the stochastic equilibrium the per-firm welfare cost is $\mathbb{E}[\mathcal{L}] = \ln \frac{\pi_0}{(r+\delta)\bar{\kappa}}$: both $\varphi$ and $\mu$ cancel. Table III summarizes the taxonomy. This map is positive, not normative: it identifies which primitive moves which margin. The welfare-improving instrument is the corrective entry tax of Section 5.2, not a change in any single primitive, and the levers below are comparative statics rather than a welfare ranking of policies.

**Growth-side and cycle-side levers are structurally separable** The taxonomy is a structural decomposition of how primitives map to outcomes. It is not a one-instrument-one-margin claim about real-world policies. Within the model, the first row of Table III is orthogonal to the others (Proposition 2). Productivity-growth interventions that move $g$ alone leave cycle frequency, amplitude, and welfare cost untouched, and cycle-side primitives

Table III: Policy anatomy: instrument-to-outcome map

| Lever | Trend | Frequency | Amplitude | Welfare cost | Example |
|---|---|---|---|---|---|
| $g$ (productivity) | ✓ | — | — | — | R&D subsidies |
| $\mu$ (matching friction) | — | ✓ | — | — | Permitting reform |
| $\varphi$ (profit curvature) | — | ✓ | ✓ | — | Cluster subsidies |
| $\bar{\kappa}$ (entry-cost level) | — | — | ✓ | ✓ | Capacity expansion ($\bar{\kappa}\downarrow$) |
| $\sigma_\kappa, \theta$ (entry-cost fluct.) | — | — | — | — | Demand stabilization |

*Notes:* ✓ marks a primitive that moves the column outcome; — marks no effect. Columns are proportional / detrended-log measures: $\omega = \sqrt{\varphi/\mu}$, mean amplitude $\sigma_x = \sqrt{\bar{b}/\varphi}$, and *cycle-induced* welfare cost $\mathbb{E}[\mathcal{L}] = \ln \frac{\pi_0}{(r+\delta)\bar{\kappa}}$ from Theorem 3. The entry-cost volatility $\sigma_\kappa$ and persistence $\theta$ (last row) move the cycle's amplitude *variability* and spectral width, dispersions the mean-amplitude column does not display, while leaving all four mean outcomes unchanged. The trend column refers to the BGP rate $g$ (Proposition 2); $g$ raises the trend on which welfare is measured (faster growth raises level welfare directly through $z_t = z_0 e^{gt}$) but does not enter the cyclical welfare cost $\mathbb{E}[\mathcal{L}]$. Welfare-cost effects are stated under the maintained per-firm metric; see Section 5.1 for scope and the leakage range under alternative metrics.

leave $g$ untouched. The decomposition gives a diagnostic: changes in cycle properties must run through $(\varphi, \mu, \bar{\kappa}, \sigma_\kappa, \theta)$, while changes in growth must run through $g$. Real-world policy bundles typically hit several primitives at once. R&D subsidies can alter $\varphi$ or the entry cost $\bar{\kappa}$ through technology channels, and workforce upgrading can raise $g$ while lowering $\mu$ or $\bar{\kappa}$. The taxonomy is therefore most useful for decomposing a given policy into margin-by-margin effects, not for predicting that any specific intervention moves only one outcome.

**CHIPS Act counterfactual** The CHIPS and Science Act bundles two structurally distinct components in the model. The first is a *growth-side* component: semiconductor R&D incentives, the National Semiconductor Technology Center, and workforce-development investments. This component raises the BGP rate $g$. The second is a *cycle-side* component: manufacturing-grant subsidies that pay for fabrication-plant construction and supplier-side capacity expansion. This component is modeled below as lowering the entry-cost level $\bar{\kappa}$, which enlarges the cycle and its welfare cost while saving on entry resources. The framework classifies these components separately. Calibrating $(\varphi, \mu, \bar{\kappa}, \sigma_\kappa, \theta)$ for NAICS 3344 (Semiconductor and Other Electronic Component Manufacturing) from BDS 1978–2022 yields a baseline spectral period $T = 4.75$ years, matching the observed BDS net-entry-rate peak. Table IV reports the model-implied effects of three stylized cycle-side interventions and one combination, all relative to baseline. The growth-side component shifts the trend without entering the table.

**Implications for the current industrial-policy debate** The model gives a nonstandard implication for CHIPS- and IRA-style entry interventions. Capacity-expansion subsidies that lower the entry-cost level $\bar{\kappa}$ *raise* per-firm welfare cost by enlarging cycle amplitude

Table IV: Illustrative policy counterfactuals, NAICS 3344

| Scenario | Period ratio | Amplitude ratio | Welfare ratio |
|---|---|---|---|
| A. CHIPS entry subsidies: $\bar{b} \times 2$ (lower $\bar{\kappa}$) | 1.00 | 1.41 | 2.00 |
| B. Cluster policy: $\varphi \times 1.2$ | 0.91 | 0.91 | 1.00 |
| C. Streamlined permitting: $\mu \times 0.7$ | 0.84 | 1.00 | 1.00 |
| A + B combined | 0.91 | 1.29 | 2.00 |

*Notes:* Calibration and full taxonomy in Online Appendix E.3. Scenario A is an illustrative entry subsidy that lowers $\bar{\kappa}$ enough to double the log-gap $\bar{b}$: cycle period unchanged, amplitude rises by $\sqrt{2}$, welfare cost doubles (exact under the log-loss metric; the proportional-loss doubling is the small-amplitude limit). Because welfare is logarithmic in $\bar{\kappa}$, the exact effective-entry-cost mapping is $\bar{\kappa}_{\text{cf}}/\bar{\kappa}_{\text{base}} = e^{-\bar{b}_{\text{base}}}$, a fractional cut of $1 - e^{-\bar{b}_{\text{base}}}$. The table reports the model-implied outcome ratios without assigning that effective-cost change a statutory subsidy rate; whether the CHIPS Act moves $\bar{\kappa}$ by the required amount is an institutional question left open. Scenario B raises $\varphi$ by 20%: cycle shortens by 9%, amplitude falls by the same factor, welfare unchanged. Scenario C reduces $\mu$ by 30%: cycle shortens by 16%, amplitude and welfare unchanged. Combined A+B speeds the cycle and raises both amplitude and welfare cost.

(Scenario A in Table IV). This result holds to leading order under the per-firm output metric. Under welfare metrics that include the supplier-side resources spent absorbing the cycle, the welfare cost of capacity expansion is attenuated, but the sign is robust (Online Appendix E.1). The recent industrial-policy turn (the CHIPS and Science Act, the Inflation Reduction Act, and the EU Net-Zero Industry Act) has revived the question of which interventions deliver welfare gains and which mainly reshape industry dynamics (Juhász et al., 2024; Rodrik, 2004; Bown, 2024). Table III sets out the full structural taxonomy. Productivity-growth interventions such as R&D, technology development, and workforce upgrading move $g$ alone, raising the BGP without affecting cycle outcomes. They should be evaluated by growth criteria. Among cycle-side levers, the entry-cost level $\bar{\kappa}$ is the only leading-order lever on the mean welfare cost; demand-stabilization lowers the entry-cost volatility $\sigma_\kappa$, reducing the cycle's variability while, at second order, slightly raising its mean cost through the convexity of the loss. Permitting reform ($\mu \downarrow$) and cluster support ($\varphi \uparrow$) speed the cycle without moving welfare. This ranking holds to leading order under the maintained per-firm metric. The qualitative ordering survives the leakage under alternative metrics (Section 5.1 and Online Appendix E.1).

# 6 Discussion

## 6.1 Modeling choices and limitations

Five limitations bound the scope of the closed-form result.

*(i) Log-quadratic exactness.* The symmetric-cycle prediction of the log-quadratic specifi-

cation is counterfactual; under general hump-shaped profits (Online Appendix B.3), cycles become asymmetric and the frequency formula is accurate to 1.6% at empirically relevant amplitudes (Online Appendix B.5 and C.2).

*(ii) Exogenous exit.* The model treats exit through the constant-rate $\delta$ rather than as a learning- or productivity-driven endogenous decision; combining timing search with Jovanovic (1982)-style learning is a natural extension.

*(iii) Linear utility and the per-firm welfare metric.* The frequency-welfare separation is exact under the maintained metric and approximately exact otherwise; the scope conditions and the leakage range under alternative metrics are stated in Section 5.1 and quantified in Online Appendix E.1.

*(iv) Within-manufacturing scope of the empirical identification.* The headline test isolates the $\varphi$ channel under the maintained restriction that $\mu$ is approximately uniform across manufacturing 3-digit industries. Section 4.3 gives the substantive treatment. Three pieces of evidence are consistent with the restriction: the recovered $\hat{\beta}_\gamma$ lies in $[0.50, 0.75]$, within one HC1 standard error of the structural 1/2; capital-side $\mu$ proxies return null coefficients; and the microfounded $\mu_i = (1-\alpha)/[\alpha(\delta_i n_{p,i})^2]$ predicts narrow within-manufacturing variation. The regulatory-side correlation is flagged, but it does not transmit to the IV slope when added as a control. A direct test of the $\mu$ channel outside manufacturing requires industry-friction-specific proxies at NAICS-3 resolution that current public data do not provide, so I leave it for future data work.

*(v) Constant-parameter benchmark vs. time-varying primitives.* The closed form $\omega = \sqrt{\varphi/\mu}$ is the *local* structural frequency under constant primitives. Slow time variation in $\varphi$ or $\mu$ can come from regulation, finance, supply-chain bottlenecks, technology, or demand composition. It generates irregular cycles disciplined by the same structural ratio. The empirical prediction then becomes a ranking statement: industries with higher persistent $\varphi/\mu$ should exhibit higher average dominant frequencies. The cross-sectional slope test in Section 4.3 captures this ranking through the persistent component of $\gamma_i$ (the 1947 historical IV) and the average spectral peak. It identifies the slope provided cross-industry variation in persistent $\varphi/\mu$ dominates transitory variation.

## 6.2 Extensions

Several extensions merit investigation: combining timing search with Jovanovic (1982)-style learning to endogenize exit; relaxing linear utility to let industries interact through aggregate consumption; endogenizing $(\varphi, \mu)$ through industry investment in linkages or entry infrastructure; combining timing search with financial frictions (Beaudry et al., 2020) to produce models with multiple spectral peaks; and full per-industry GMM/SMM estimation of $(\varphi_i, \mu_i, \bar{\kappa}_i, \theta_i, \sigma_{\kappa,i})$ with overidentification testing (the three-moment fingerprint of Section 4.4 is a structural-implication test, not a structural-estimation exercise).

# 7 Concluding remarks

This paper proposes a structural account of why different industries fluctuate at different frequencies. The central claim is that cycle length is a property of industry market structure: the curvature of the profit landscape and the friction of entry matching. It is not only a property of the shock process the industry inherits. Under *timing search*, agents choose *when* to enter a market subject to matching frictions that penalize rapid adjustment. The equilibrium indifference condition across entry dates forces overshoot rather than monotone adjustment toward optimal scale. This generates a cycle whose frequency $\omega_0 = \sqrt{\varphi/\mu}$ is pinned by two market-structure primitives: the profit-landscape curvature $\varphi$ and the entry-matching friction $\mu$. The same logic could be extended to other timing decisions, including capital investment, hiring, technology adoption, and financial-market participation.

U.S. industry data support this prediction. Across 15 manufacturing 3-digit industries, industries with stronger agglomeration have higher spectral peak frequencies. The cross-industry slope linking concentration to cycle frequency is close to the structural value. The pattern is consistent across alternative spectral estimators and survives instrumental-variable identification using historical industrial geography. Standard business-cycle benchmarks can reproduce cross-industry frequency dispersion, but they do not on their own deliver this slope. The data align with the timing-search account.

Beyond the cross-industry frequency prediction, the model implies that cycle frequency and the welfare cost of cycles are governed by disjoint sets of market primitives. Market structure pins where in the spectrum an industry oscillates. A separate primitive, the entry-cost level, pins how costly those oscillations are. This separation gives a policy diagnostic for entry-side industrial policy. Streamlining entry matching and strengthening agglomeration change cycle speed without moving welfare cost; the welfare cost itself reflects a congestion externality in the timing of entry that a rebated corrective tax can address, while capacity-expansion subsidies enlarge the cycle (Sections 5.2–5.3). In this view, an industry's cycle frequency depends as much on its market structure as on the shocks it receives.

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